BorBann
A Real Estate Information Platform

by

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Submitted as
Software Requirements Specification
for
01219395 Innovative Software Group Project Preparation

Faculty of Engineering
Kasetsart University
Bangkok, Thailand

March 2026

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This Project Submitted in Partial Fulfillment of the Requirements for Bachelor Degree of Engineering (Software Engineering)

Department of Computer Engineering, Faculty of Engineering
KASETSART UNIVERSITY
Academic Year 2025

Approved by:

Advisor .... Date.../.../...
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Abstract

BorBann is a real estate information platform developed to support homebuyers, investors, and business users in Bangkok. The project addresses a common problem in the Thai real estate market: important information is spread across many sources and is difficult to compare in one place. To solve this, the platform combines property data, public records, points of interest, transit information, and environmental risk indicators in a single web-based system.

The delivered system includes an interactive map, a location intelligence dashboard, an explainable property valuation module, and an AI chat assistant that can answer location-specific questions through retrieval and tool-based reasoning. The frontend is implemented with React, while the backend uses FastAPI, PostgreSQL, and PostGIS for geospatial processing. For valuation, the current production path uses a LightGBM model with spatial features, while graph-based models are retained as an experimental direction.

The project also places emphasis on trust and verification. The platform provides confidence signals, feature-based explanations, score legends, and source tooltips so that users can understand both the result and its origin. Evaluation covers functional testing, model benchmarking, explainability checks, chatbot quality checks, and load testing. Initial feedback from a small Google Form study with five users shows that users value the map-based workflow, readable summaries, and localized insights, while also requesting stronger chatbot accuracy and clearer visual cues in some views. Overall, BorBann demonstrates that a locally adapted, data-driven platform can improve the accessibility and usefulness of real estate information in Thailand.

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Chapter 1

Introduction

1.1 Background

The global real estate market shows positive growth trends despite various challenges. Real estate market remains a key sector for people like homebuyers and investors, influenced by complex characteristics with heterogeneous nature and affected by numerous elements like policies and consumer trends. The market is projected to grow from $4,143.71 billion in 2024 to $4,466.58 billion in 2025, and global real estate investment volumes continue to increase in 2024[1].

Thailand's real estate market faces both challenges and opportunities in the current economic climate. The real estate market in Thailand, especially in Bangkok in 2024-2025, faces challenges such as high household debt, strict credit conditions, and rising costs causing market contraction, decreasing project launches by 43.72% in 2024[2]. However, it also presents opportunities for recovery through foreign investment driven by tourism recovery, government incentives, and technological advancements. Condominiums remain attractive for rental yields and foreign buyers, while the housing market has moderate growth supported by infrastructure development.

Crucially, these market opportunities can significantly undermined by Thailand's real estate information ecosystem, which suffers from several structural limitations, including outdated listings, lack of official transaction data, and data fragmentation[3]. Addressing these issues is essential for maximizing Thailand's market potential and enabling effective decision-making for both homebuyers and investors seeking to capitalize on the available opportunities.

1.2 Problem Statement

Even with large amounts of real estate data available through various channels including property listings, historical transaction records, and news, investors still face challenges in aggregating, contextualizing, and deriving action from these sources. Additionally, the Thai real estate market has different characteristics from other countries because of the limited availability of official property transaction records, listing duplication across platforms, and less standardized data compared to more mature markets.

Current platforms in Thailand like DDProperty, Hipflat, and Baania lack many important contextual factors such as climate risk assessments, neighborhood-specific news, and local beliefs that influence property valuation and real estate investment in the long term.

While platforms like Zillow and House Canary have advanced and comprehensive real estate analytics, predictive analytics, and user-friendly interfaces, they do not operate in Thailand. Even if these advanced platforms were to enter the Thai market, they would face challenges adapting their algorithms to the local context. Their models are calibrated to markets with standardized property classifications, valuations that vary by developers, and consistent transaction data—elements that are limited in Thailand's real estate ecosystem.

The BorBann platform addresses these challenges and aims to help users by creating a real estate data platform integrated with artificial intelligence, geospatial analytics, and data aggregation by focusing on analytics rather than transaction facilitation.

1.3 Solution Overview

BorBann will function as real estate data platform to users, integrating multiple data sources with advanced analytics. Below are the features and their details.

Borbann AI Agent (Agentic Workflow)

- **Natural Language Interface:** Chat with the system to query locations ("How is the flooding here?")
- **Tool Orchestration:** Automatically selects and executes the right tools (e.g., transit scoring, risk checking) based on user intent
- **Spatial Context Awareness:** Understands user interactions like dropping pins or drawing bounding boxes

Site Viability & Catchment Analysis

- **Commercial Site Scoring:** Evaluate business potential (e.g., for a Cafe) based on traffic magnets vs. competitors
- **Isochrone Visualization:** Visualizing 15-minute walking/driving reach from any point
- **Livability Scores:** Composite scoring for Transit, Walkability, and Schools

Local Contextual Analytics

- **Environmental risk assessment:** Evaluate flood risk and noise pollution levels
- **Facility proximity analysis:** Calculate accessibility to schools, transit, and commercial centers
- **Neighborhood quality scoring:** Generate composite metrics for area evaluation

Explainable Price Prediction Model

- **Feature importance analysis:** Quantify and rank factors influencing property prices
- **Adjustable characteristics modeling:** Adjust property characteristics to visualize price impacts
- **Confidence intervals:** Provide lower/upper price bounds for realistic expectations
- **Factor categorization:** Group influences by type (property features, location, market trends)
- **Natural language explanations:** Generate readable summaries of price determinants
- **Visual breakdowns:** Display contribution percentages and relationship graphs

Geospatial Visualization

- **Heatmap generation:** Create density visualizations for environmental factors, pricing, and metrics
- **Geospatial analytics:** Calculate analytics for custom geographic areas

1.4 Target User

User Type	Description	Needs
Real Estate Investors	Individuals focused on maximizing long-term investment, including foreign investors	• Investment analysis • Supporting data for decision making
Homebuyers	First-time purchasers, residents looking to relocate within Thailand, and expats seeking housing	• Property comparisons • Neighborhood insights • Pricing guidance

Table 1.1: Target Users and Their Needs

1.5 Benefit

The BorBann platform will provide numerous benefits to the Thai real estate market. It improves market transparency by enhancing accessibility to market information. It also helps both homebuyers and investors reduce research time, achieve lower transaction risks, and discover overlooked investment opportunities. Additionally, the platform will effectively represent the unique characteristics of the Thai real estate market.

1.6 Terminology

Term	Definition
Local Analytics	Analysis focused on extremely specific geographic areas, such as neighborhoods or even individual streets, to provide highly relevant insights.
Price Prediction Model	An algorithm or statistical model that forecasts property values based on historical data, market trends, and various property characteristics.
Proximity Analysis	The study of spatial relationships between geographic features, typically to evaluate the distance between properties and amenities or services.
Geospatial Visualization	The graphical representation of data with a geographic or spatial component, often through maps and interactive displays.

Table 1.2: Terminology

Chapter 2

Literature Review and Related Work

This chapter presents a review of existing platforms in the domain of real estate analytics and information systems. This review includes both international and Thai platforms, their features, strengths, and limitations. The analysis establishes the current state of real estate information platforms and identifies opportunities for the BorBann platform to address unmet needs in the Thai market.

2.1 Competitor Analysis

Many real estate platforms primarily function as property listing aggregators rather than comprehensive information systems. Their features support transaction facilitation through showcasing available properties, while offering limited analytical tools for market understanding. However, platforms like House Canary represent exceptions, operating specifically as information systems that provide investors with data analytics and market insights to support evidence-based decision-making in real estate investments.

Feature	BorBann (Proposed)	DD Property	Hipflat	House Canary	Zillow
Borbann AI Agent (ReAct)	**Yes**	No	No	No	No
Site Viability & Catchment Analysis	**Yes**	No	No	No	No
Local Contextual Analytics	**Yes**	No	No	Yes (Not optimized for Thailand)	Yes (Not optimized for Thailand)
Explainable Price Prediction Model	**Yes**	No	No	No	No
Geospatial Visualization	**Yes**	Yes	Yes	Yes	Yes

Table 2.1: Feature Comparison: BorBann vs Other Platforms

Table 2.1 demonstrates BorBann's technical advantages in the real estate analytics market. While all platforms offer geospatial visualization, BorBann's implementation includes advanced analytics like climate assessment, matching international platforms but surpassing local Thai competitors. BorBann's AI Agent allows natural language interaction for complex queries, unlike traditional filter-based search. For local contextual analytics, BorBann provides Thailand-optimized insights including weather patterns and population density, whereas DDProperty/Hipflat only show basic nearby facilities, and international platforms lack Thailand-specific optimization. Most distinctively, BorBann's price prediction model prioritizes explainability and interpretability, revealing the reasoning behind valuations rather than presenting opaque predictions like competing platforms.

Technical Aspect	DD Property	Hipflat	House Canary	Zillow	BorBann (Proposed)
Data Sources	User-submitted, Proprietary	User-submitted, Proprietary	Proprietary	Multiple Sources, Proprietary	Open Data, User-submitted
ML Implementation	Basic Prediction	None	Black-box Models	Black-box Models	Explainable Models

Table 2.2: Comprehensive Technical Comparison

Table 2.2 highlights key technical differences between BorBann and competing platforms. For data sources, while competitors rely heavily on government or user-submitted data, BorBann uniquely leverages open data combined with APIs to build a more comprehensive dataset. Regarding machine learning, BorBann distinguishes itself by implementing explainable models, providing transparency in its predictions. This contrasts with DDProperty's basic prediction capabilities, Hipflat's complete lack of ML features, and the black-box approaches of House Canary and Zillow where prediction logic remains hidden from users.

Chapter 3

Requirement Analysis

3.1 Stakeholder Analysis

The stakeholder landscape for BorBann includes primary stakeholders who directly interact with the platform and secondary stakeholders who indirectly influence its adoption and effectiveness. Recognizing these groups' specific needs helps develop a platform that delivers value across the real estate ecosystem.

Stakeholder Category	Stakeholder Group	Key Interests	Primary Requirements
Primary	Real Estate Investors	Environment risk assessment	Advanced analytics, predictive modeling, risk scoring
Primary	Homebuyers	Affordability, area quality, lifestyle alignment, environmental factors	User-friendly interface, neighborhood insights, price comparisons, risk assessments
Secondary	Real Estate Agencies	Market positioning, client advisement, property valuation	Property data, reliable market insights

Table 3.1: BorBann Platform Stakeholder Summary

The primary stakeholders are direct users who rely on BorBann's capabilities for informed real estate decisions. Their requirements range from investment analytics to intuitive neighborhood quality assessments. Secondary stakeholders like real estate agencies can influence platform adoption and serve as valuable data partners.

Primary Stakeholders

Primary stakeholders are those who directly use or are significantly affected by the BorBann platform. They rely on its capabilities to make informed decisions about real estate investments, purchases, and market trends.

Real Estate Investors

Real estate investors use BorBann to identify profitable investment, assess risks, and make data-driven decisions.

Interests and Concerns:

- Maximizing return on investment
- Identifying growth areas
- Risk assessment
- Diversification across neighborhoods and property classes

Requirements:

- Advanced analytics tools that is customizable
- Predictive price modeling with easy to understand explaination
- Risk scores in each area based on multiple factors

Homebuyers

Homebuyers rely on BorBann to find properties that align with their budget, lifestyle, and long-term needs. The platform helps them assess affordability, neighborhood quality, and potential risks.

Interests and Concerns:

- Property affordability and financing options
- Area quality
- Location amenities and lifestyle alignment
- Environmental factors including flood and pollution risks
- Commute time to work, schools, and essential services
- School districts and educational quality

Requirements:

- Intuitive, user-friendly interface with minimal learning curve
- Comprehensive neighborhood insights including safety metrics
- Price comparison tools with historical context

- Property quality and developer reputation metrics
- Detailed flood risk and environmental quality assessment
- Transit accessibility maps with time-based visualizations
- Education quality indicators and school zone mapping

Secondary Stakeholders

Secondary stakeholders are indirectly impacted by the BorBann platform, influencing its adoption, data availability, and overall market reach.

Real Estate Agencies

Real estate agencies act as intermediaries between property buyers and sellers. They use BorBann to improve their advisory services, gain a competitive edge, and provide market insights to clients.

Interests and Concerns:

- Market positioning relative to competitors
- Client advisement based on reliable data
- Access to comprehensive property information
- Accurate property valuation to support transactions

Influence:

- Potential data partners and platform promoters
- Can significantly influence adoption rates among clients
- May provide valuable transaction data not available elsewhere

3.2 User Stories

User stories capture the essential needs and goals of the BorBann platform's target users from their perspective. These stories follow the standard format: "As a [user type], I want to [action/feature] so that [benefit/value]."

Table 3.2: User Stories with Acceptance Criteria

User Story	Acceptance Criteria
Local Contextual Analytics	
As a user, I can see all contextual data for specific areas to make informed decisions	• System displays at least 5 contextual data points for any selected area
As a user, I want to see flood risk assessments for properties with historical flooding data	• Flood risk presented on a 5-point scale with historical context • System shows flood history for the past 10 years when available
As a user, I want to see all schools within a custom radius, including distance, ratings, and types	• System displays all schools within user-defined radius • Each school listing includes distance, type, and quality rating
As a user, I want to view healthcare facilities near a property with distance and service information	• Healthcare facilities categorized by type (hospital, clinic, etc.) • Distance and basic service information provided for each facility
Explainable Price Prediction Model	
As a user, I want to see how specific contextual factors influence the property's predicted price	• System displays top 5 factors influencing price with percentage contribution • Visual indicators show positive/negative impact of each factor
As a user, I want the model to be interpretable so I can understand factors affecting the price	• Plain language explanations accompany each prediction • Interactive elements allow exploration of factor relationships

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Table 3.2 – Continued from previous page

User Story	Acceptance Criteria
As a user, I want a statement or reason to back the prediction so I can trust the system's valuation	• Each prediction includes at least 3 specific supporting statements • System indicates confidence level for each prediction
As a user, I want to see a predicted price range for any property I select	• System shows lower and upper bounds for predicted price
As a user, I want to understand how the model derives the result so I can explain the valuation to others	• System provides visual breakdown of prediction process
Geospatial Visualization	
As a user, I can see property listings on the map and click them to view detailed information	• Property details popup appears within 1 second of clicking a marker • Popup contains at least 5 key property attributes
As a user, I can see sections of the same property group to identify properties from the same development	• Properties from same development visually grouped with distinct boundaries • Group name appears when hovering over grouped properties
As a user, I can see multiple map visualization types to analyze different environmental factors	• System offers at least 5 different visualization overlays • Users can toggle between visualizations without page reload
As a user, I want to pan and zoom on an interactive property map to explore different areas efficiently	• Map responds to standard pan/zoom gestures within 100ms • Property markers adjust density based on zoom level

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User Story	**Acceptance Criteria**
As a user, I want to set a custom radius around a point on the map to analyze the surrounding area	• User can place and adjust analysis radius on any map location
Site Analysis & Catchment	
As a business owner, I want to see a site score for a specific category (e.g., Cafe) to evaluate commercial potential	• System provides a 0-100 score based on magnets and competition • Analysis includes count of traffic drivers within 1km
As a user, I want to visualize the 15-minute reachable area from a location to understand accessibility	• Map displays an isochrone polygon for walking or driving • Polygon overlay includes estimated population reach
Borbann AI Agent	
As a user, I want to ask natural language questions about properties and have the system execute the right tools to answer	• Agent breaks down complex queries into steps (e.g., search, then calculate score) • UI shows the "thought process" and tools being used
As a user, I want to drag and drop a pin or draw a box on the map to define the search area for the agent	• Agent recognizes the spatial context (coordinates or bounding box) from the UI action • Search results are constrained to the defined area

3.3 Use Case Diagram

The Use Case Diagram for the BorBann platform, shown in Figure 3.1, illustrates the primary interactions between the system and its three main user types: Real Estate Investors, Homebuyers, and Property Developers. The diagram captures the core functionality of the platform and how different users interact with its features.

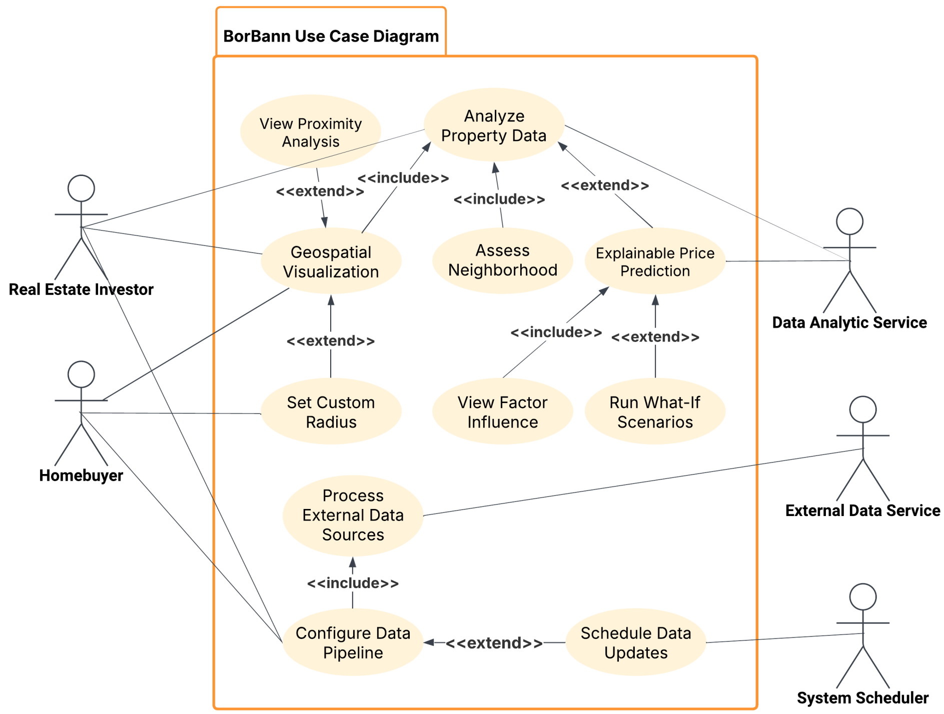


Figure 3.1: Use Case Diagram

Primary Actors

The use case diagram identifies five key actors interacting with the BorBann system:

- **Real Estate Investor** Seeks in-depth property analytics, market trends, and investment decision support.
- **Homebuyer** Interested in residential properties, lifestyle factors, and long-term value.
- **Data Analytic Service** Provides analytic capabilities to support prediction and data analysis features.
- **External Data Service** Supplies third-party data inputs via APIs and other integrations.

- **System Scheduler** Automates routine system tasks such as scheduled data refresh operations.

Core Use Cases

The BorBann platform provides the following primary functionalities:

- **Analyze Property Data** Allows users to examine detailed property information, metrics, and comparisons.
  - *Includes:* Assess Neighborhood
  - *Extended by:* Explainable Price Prediction
- **Geospatial Visualization** Interactive map-based visualization of property and neighborhood data.
  - *Extended by:* Set Custom Radius, View Proximity Analysis
- **Assess Neighborhood** Evaluates contextual factors around properties (e.g., schools, amenities).
  - *Includes:* View Factor Influence
  - *Extended by:* Check Environmental Risks
- **Explainable Price Prediction** AI-generated pricing insights with interpretable influencing factors.
  - *Includes:* View Factor Influence
- **Interact with AI Agent** Enables natural language queries and spatial context injection (pins/bbox) to orchestrate tools.
  - *Includes:* Site Analysis, Property Search

Extended Functionality

The following optional use cases extend the platform's functionality:

- **View Proximity Analysis** Triggered when users want detailed proximity-based data (e.g., nearby schools, transport).
- **Set Custom Radius** Adds user-defined range settings for map-based filtering and analysis.
- **Check Environmental Risks** Enables users to access data on environmental threats (e.g., flood zones, pollution).

Actor-System Interactions

Each actor interacts with the system in distinct ways, as shown in the diagram:

- **Real Estate Investor**
  - Initiates *Analyze Property Data*, *Geospatial Visualization*, and *Configure Data Pipeline*
  - Benefits from extended insights like *Set Custom Radius*, *View Proximity Analysis*, and *Explainable Price Prediction*
- **Homebuyer**
  - Accesses *Analyze Property Data*, *Geospatial Visualization*, *Assess Neighborhood*, and *Explainable Price Prediction*
  - Makes use of neighborhood-specific features like *View Factor Influence* and *Check Environmental Risks*
- **Data Analytic Service**
  - Collaborates with the system to enable *Explainable Price Prediction*
- **External Data Service**
  - Supports *Process External Data Sources* as part of the data ingestion workflow
- **System Scheduler**
  - Triggers *Schedule Data Updates* as an automated background process

Use Case Relationships

- **<<include>>** Represents mandatory sub-functions:
  - *View Factor Influence* is included in both *Assess Neighborhood* and *Explainable Price Prediction*
  - *Process External Data Sources* is included in *Configure Data Pipeline*
- **<<extend>>** Represents optional or conditional behavior:
  - *Set Custom Radius* and *View Proximity Analysis* extend *Geospatial Visualization*
  - *Check Environmental Risks* extends *Assess Neighborhood*
  - *Schedule Data Updates* extends *Configure Data Pipeline*

3.4 Use Case Model

The Use Case Model details the interactions depicted in the Use Case Diagram, describing each use case within BorBann's core features.

Use Case Name	**View Property Insights**
Actors	Real Estate Investor, Homebuyer, Property Developer
Description	Provides users with comprehensive analytics and contextual information about specific properties
Preconditions	User has selected a property from search results
Basic Flow	1. User selects a property for detailed viewing 2. System retrieves comprehensive property data 3. System presents property details with analytics 4. System displays neighborhood characteristics 5. System shows historical performance metrics 6. User reviews information
Alternative Flows	• User can extend to save the property as a favorite • User can extend to view explainable price predictions • User can request additional specific analytics
Postconditions	User gains comprehensive insights about the property
Associated Feature	Local Contextual Analytics - providing trend analysis and neighborhood-specific insights

Table 3.3: View Property Insights Use Case

Use Case Name	Explainable Price Prediction
Actors	Real Estate Investor, Homebuyer
Description	Provides transparent, interpretable price predictions with detailed explanations of contributing factors
Preconditions	User is viewing property insights
Basic Flow	1. User requests price prediction for a property 2. System calculates predicted price using its models 3. System generates explanation of factors influencing the prediction 4. System presents prediction with confidence interval 5. System displays factor weights and their impact 6. User reviews prediction and explanation
Alternative Flows	• User can adjust factors to see impact on prediction • User can compare prediction with market averages • User can save prediction for future reference
Postconditions	User understands the predicted price and the reasoning behind it
Associated Feature	Explainable Price Prediction Model - providing transparent explanations for price predictions

Table 3.4: Explainable Price Prediction Use Case

3.5 User Interface Design

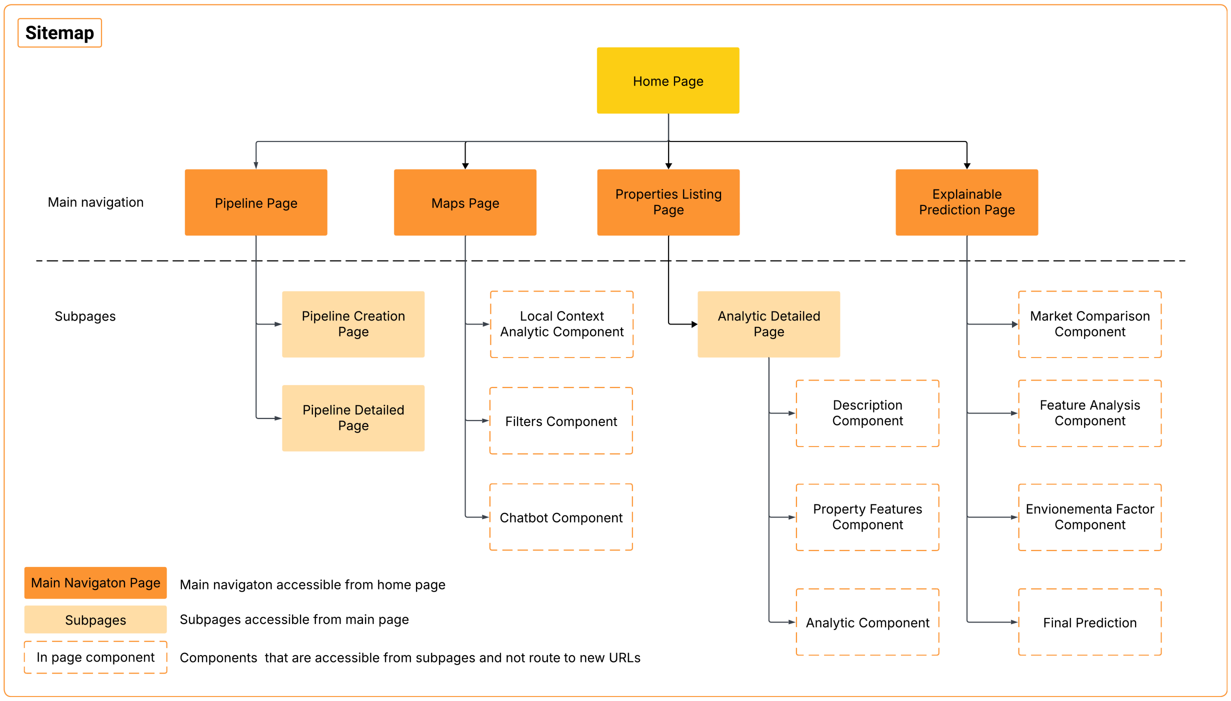


Figure 3.2: Sitemap - Platform Structure Overview

Figure 3.2 shows the structure of the BorBann platform. The Home Page acts as the central access point leading to four main sections: Pipeline, Maps, Properties Listing, and Explainable Prediction pages.

Each main section connects to specific subpages. The Pipeline section includes Creation and Detailed pages. The Maps section features Local Context Analytics, Filters, and Chatbot components. The Properties section provides Analytic Detailed pages with Description, Property Features, and Analytics components. The Explainable Prediction section offers Market Comparison, Feature Analysis, Environmental Factor analysis, and Final Prediction components.

This structure organizes the platform's key functions in a logical flow, making it easy for users to navigate between data management, visualization, and prediction features.

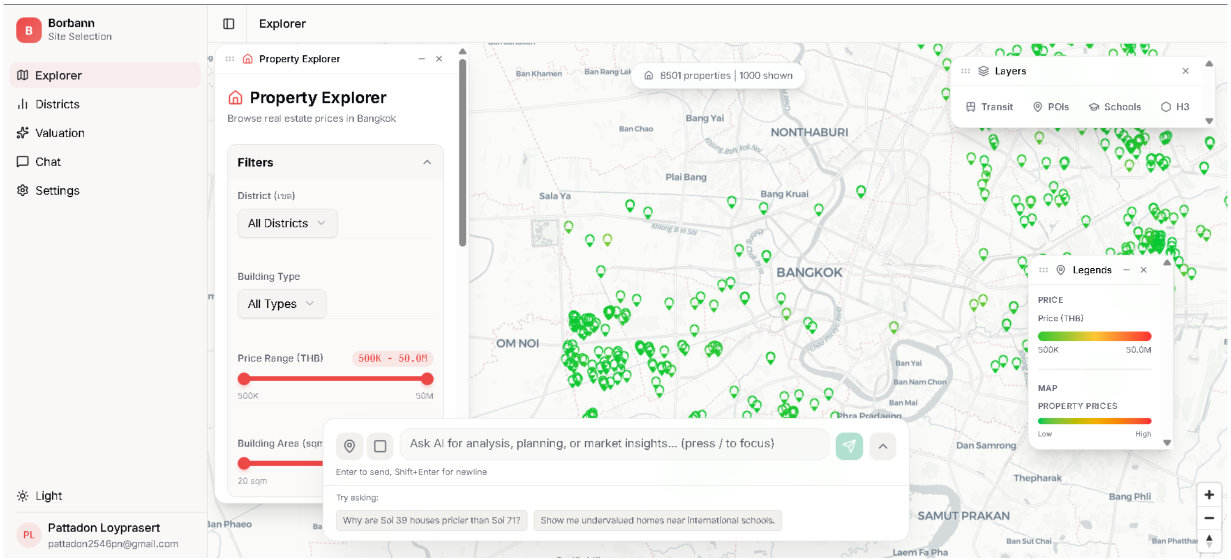


Figure 3.3: BorBann Platform Homepage

Figure 3.3 shows the BorBann platform homepage, which provides an intuitive entry point to the system. The page presents the core value proposition with "Property Analytics Simplified" and briefly explains how BorBann helps users make data-driven real estate decisions through analytics, geospatial visualization, and market insights. Two primary action buttons - "Explore Map" and "Browse Properties" - enable quick access to key functionality. The lower section showcases three main analytics tools: Geospatial Visualization for interactive property mapping, Price Prediction with explainable AI features, and Data Pipeline for automated data collection and processing.

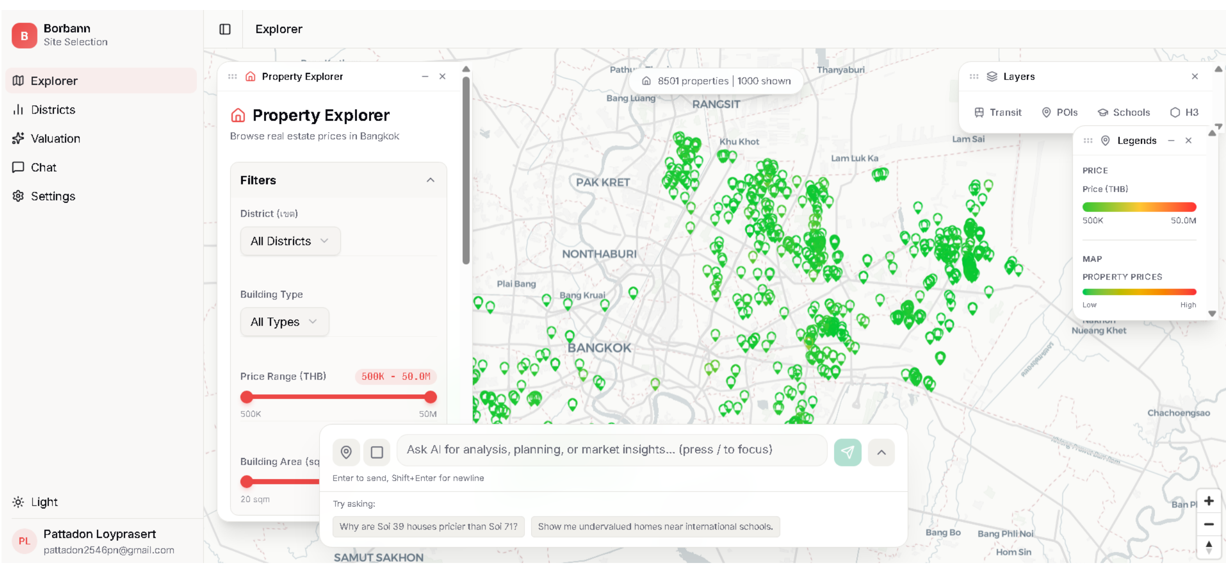


Figure 3.4: Geospatial Visualization - Main Map Interface

Figure 3.4 displays the interactive property map with color-coded markers and navigation controls. The interface supports pan/zoom gestures and provides filtering options through the sidebar.

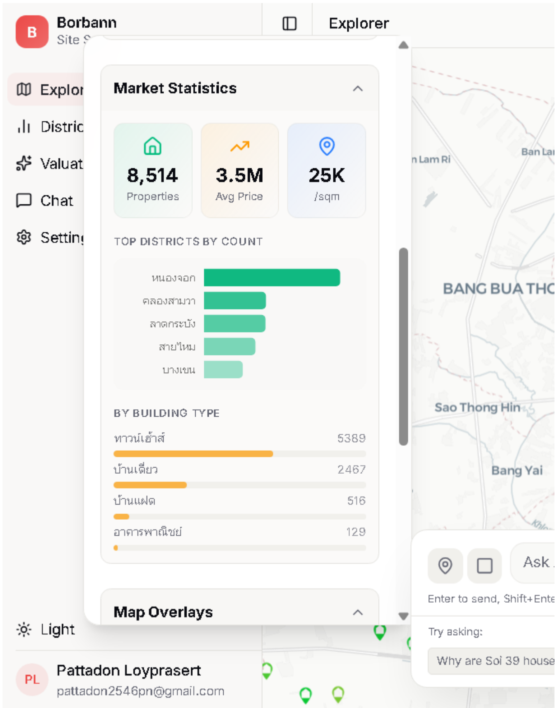


Figure 3.5: Map Analytics Overlay

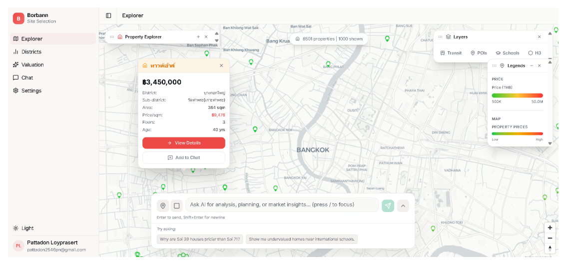


Figure 3.6: Property Detail Overlay

Figure 3.5 shows environmental factor visualization through heat maps and data layers. This overlay helps users analyze contextual factors like pollution levels directly on the map.

Figure 3.6 displays the popup that appears when users click map markers. This overlay provides immediate property information without requiring navigation away from the map.

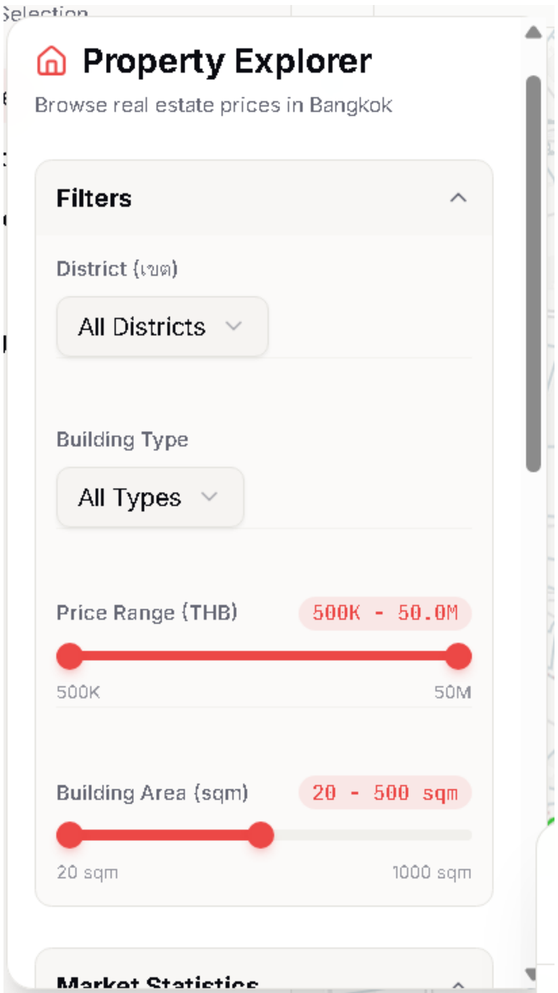


Figure 3.7: Property Filter Panel

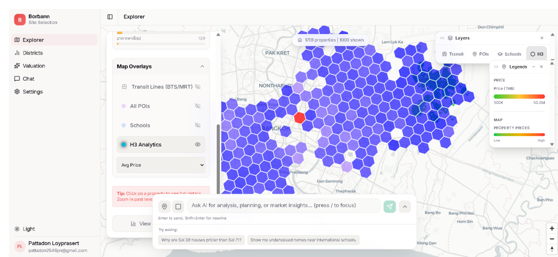


Figure 3.8: Advanced Property Filter Options

Figure 3.7 shows basic property filtering controls for price range, type and size parameters. The panel uses sliders and checkboxes for intuitive refinement of map results.

Figure 3.8 displays extended filtering options for specific amenities and neighborhood characteristics. These advanced parameters enable precise property matching based on detailed criteria.

Figure 3.9 shows the conversational assistant for property search and analysis. The chatbot handles natural language queries about properties and market conditions.

Figure 3.10 displays the initial analysis of property attributes in the prediction model. The interface includes interactive parameter sliders that show how changes to property characteristics affect the predicted price.

Figure 3.11 shows the contribution of different property features to the predicted price. The visualization uses color-coded bars to differentiate positive and negative factors with percentage impact values.

Figure 3.12 presents a comparison of the subject property against similar properties in

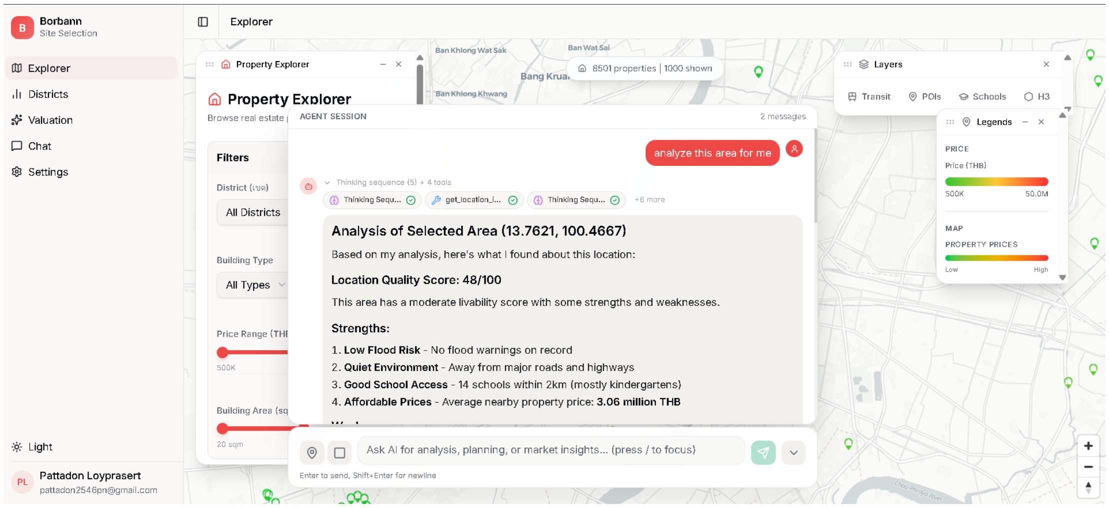


Figure 3.9: Interactive Chatbot Assistant

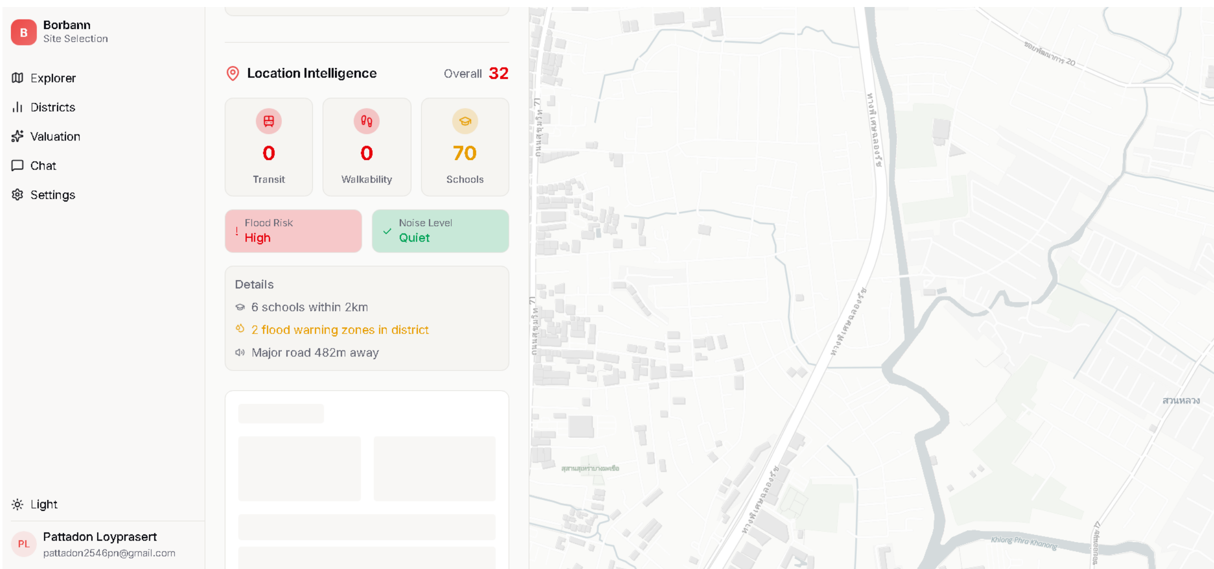


Figure 3.10: Explainable Price Prediction - Property Overview

the area. The table highlights key attributes and pricing metrics with supporting analysis of market positioning.

Figure 3.13 analyzes environmental conditions and nearby amenities affecting property value. The interface displays flood risk, air quality, and proximity to key facilities with quantified impact on valuation.

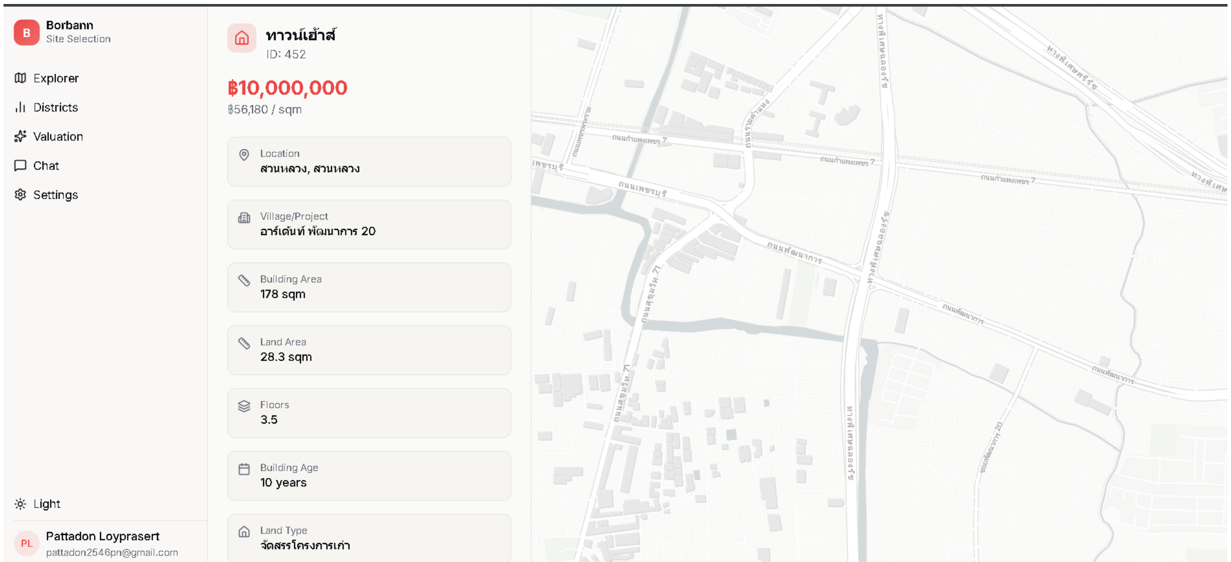


Figure 3.11: Explainable Price Prediction - Feature Importance Visualization

Figure 3.14 shows the final price prediction with confidence metrics and range indicators. The summary section explains how the valuation synthesizes insights from all analytical components.

Figure 3.15 display property listings page with all listing that follow the filter tab the right.

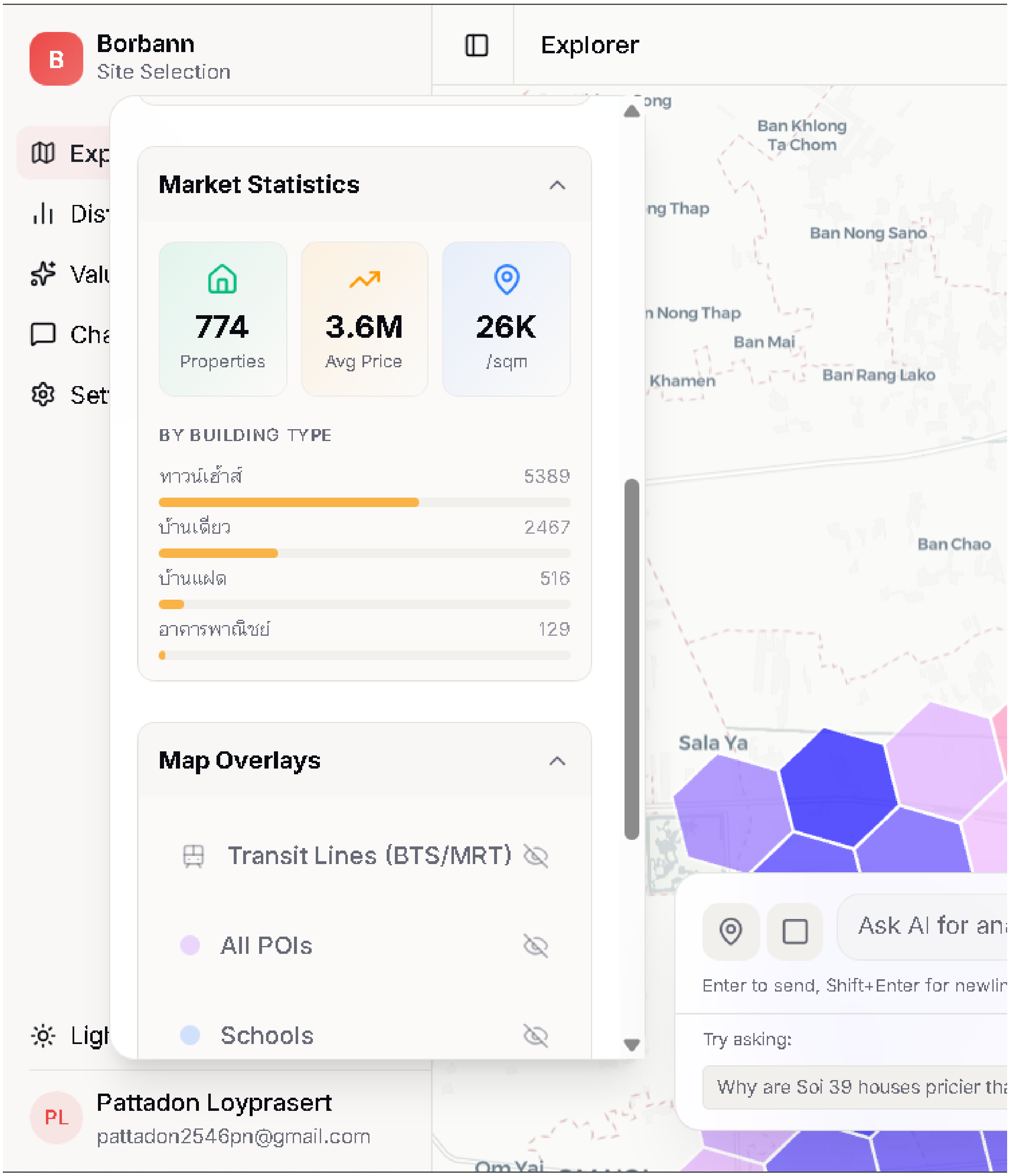


Figure 3.12: Explainable Price Prediction - Comparative Market Analysis

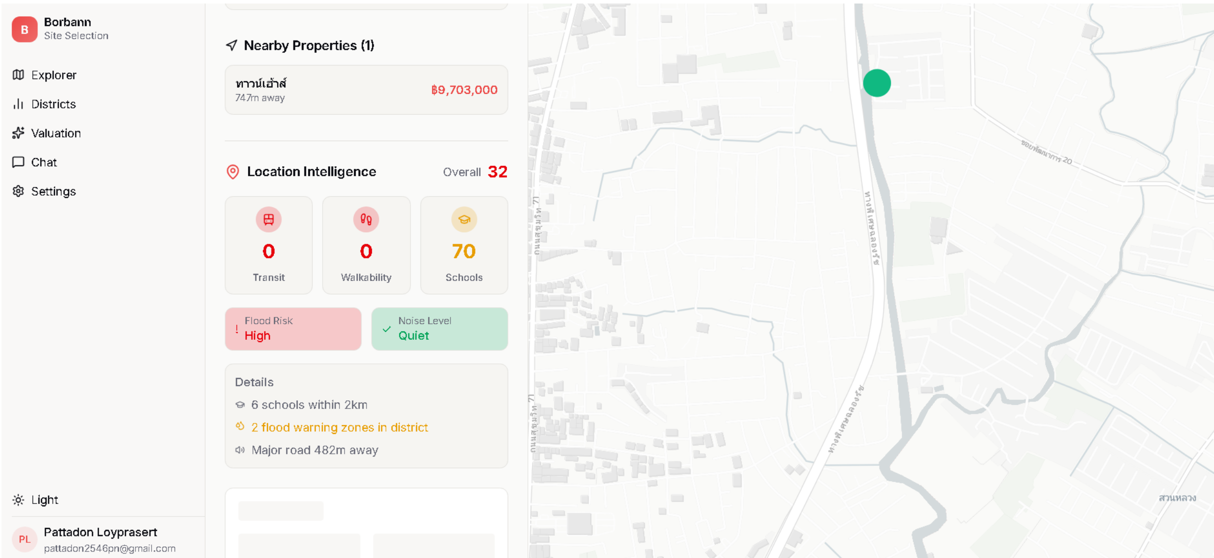


Figure 3.13: Explainable Price Prediction - Environmental Impact Analysis

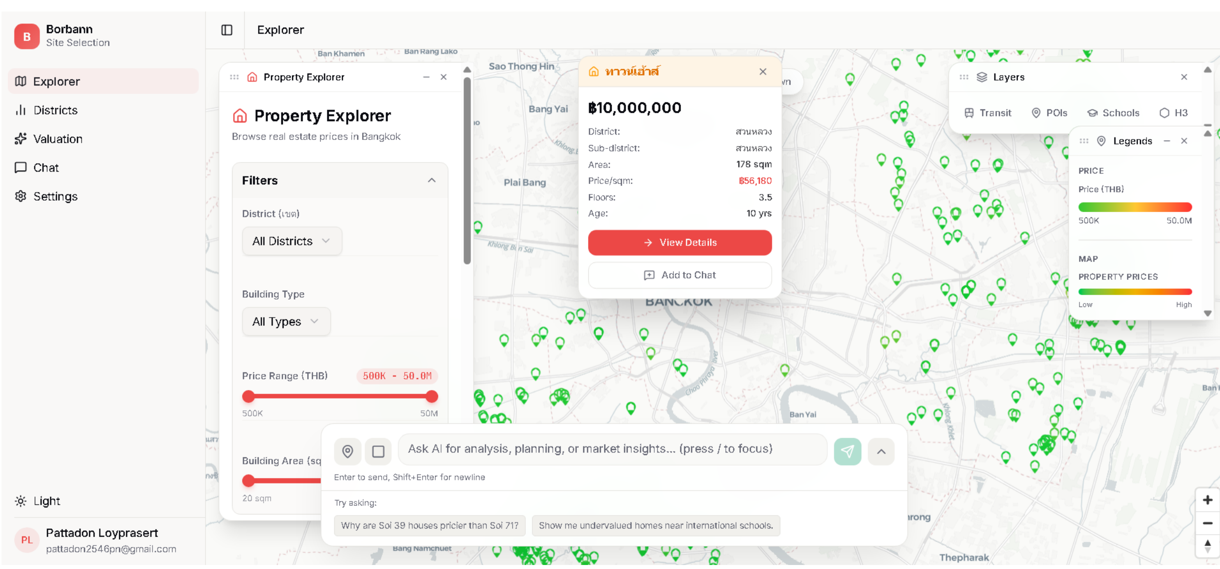


Figure 3.14: Explainable Price Prediction - Final Valuation with Confidence Range

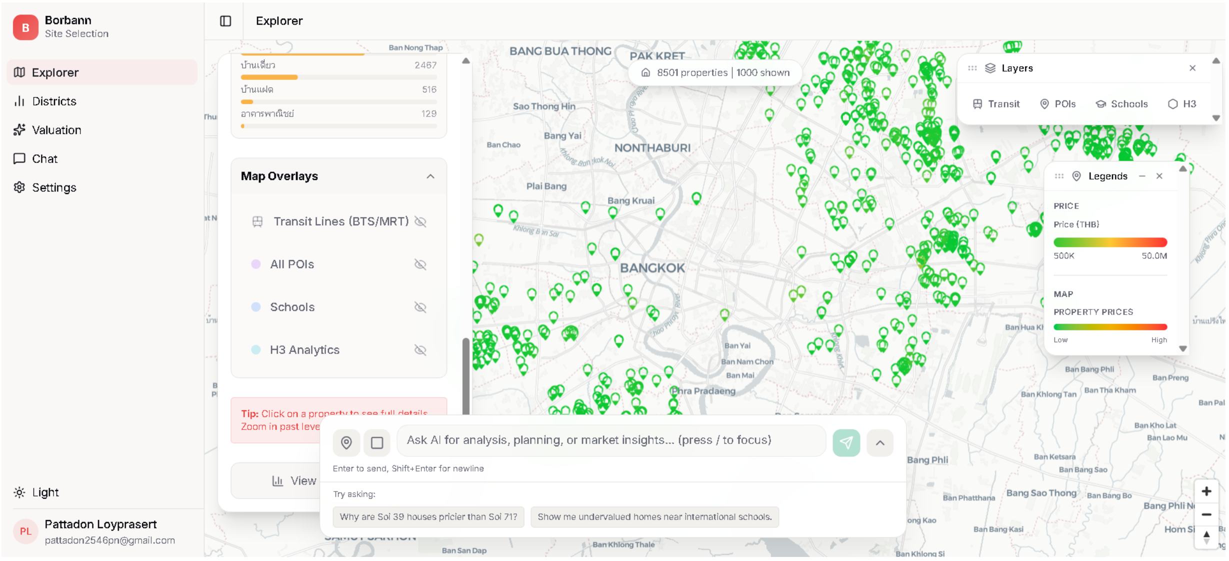


Figure 3.15: Property Listings Page

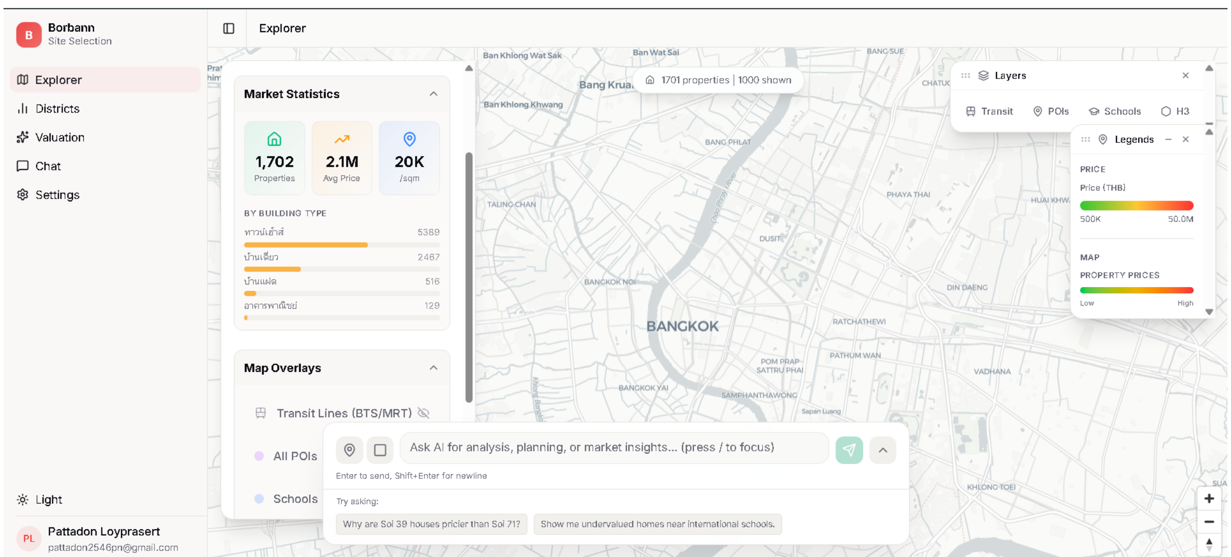


Figure 3.16: Property Listings Page - Analytic Tab

Figure 3.16 shows the analytic tab that show local context analytic of that specific listing.

Chapter 4

Software Architecture Design

This chapter presents the core software-design artifacts used to describe the BorBann platform. The purpose of this chapter is to show the structural and behavioral design of the system before implementation details are discussed in later chapters. The design is presented through four complementary views: the domain model, the design class diagram, the sequence diagram, and the overall system architecture.

4.1 Domain Model

The domain model describes the main problem-space entities involved in the BorBann platform and the relationships among them. In this project, the domain includes real-estate properties, user sessions, valuation requests, geospatial context, AI-assisted interactions, and supporting analytic outputs. These concepts define the information structure required to support property exploration, valuation, and conversational assistance.

Figure 4.1 illustrates the domain model used in the system design.

The model centers on the property as the primary business entity. A property is associated with location attributes, valuation-related features, market context, and user interactions. Users may search for properties, request valuations, inspect neighborhood intelligence, and interact with the conversational agent. These actions produce derived outputs such as price estimates, explanation factors, comparable-property summaries, and location-intelligence scores. The domain model therefore captures both persistent business entities and transient analysis artifacts that are required for the platform to function as an intelligent decision-support system.

4.2 Design Class Diagram

The design class diagram translates the domain concepts into software-level structures. It shows the principal classes, services, and interface boundaries used to organize the implementation. In BorBann, the design emphasizes modularity between frontend interaction, backend routing, AI services, geospatial services, and persistence layers.

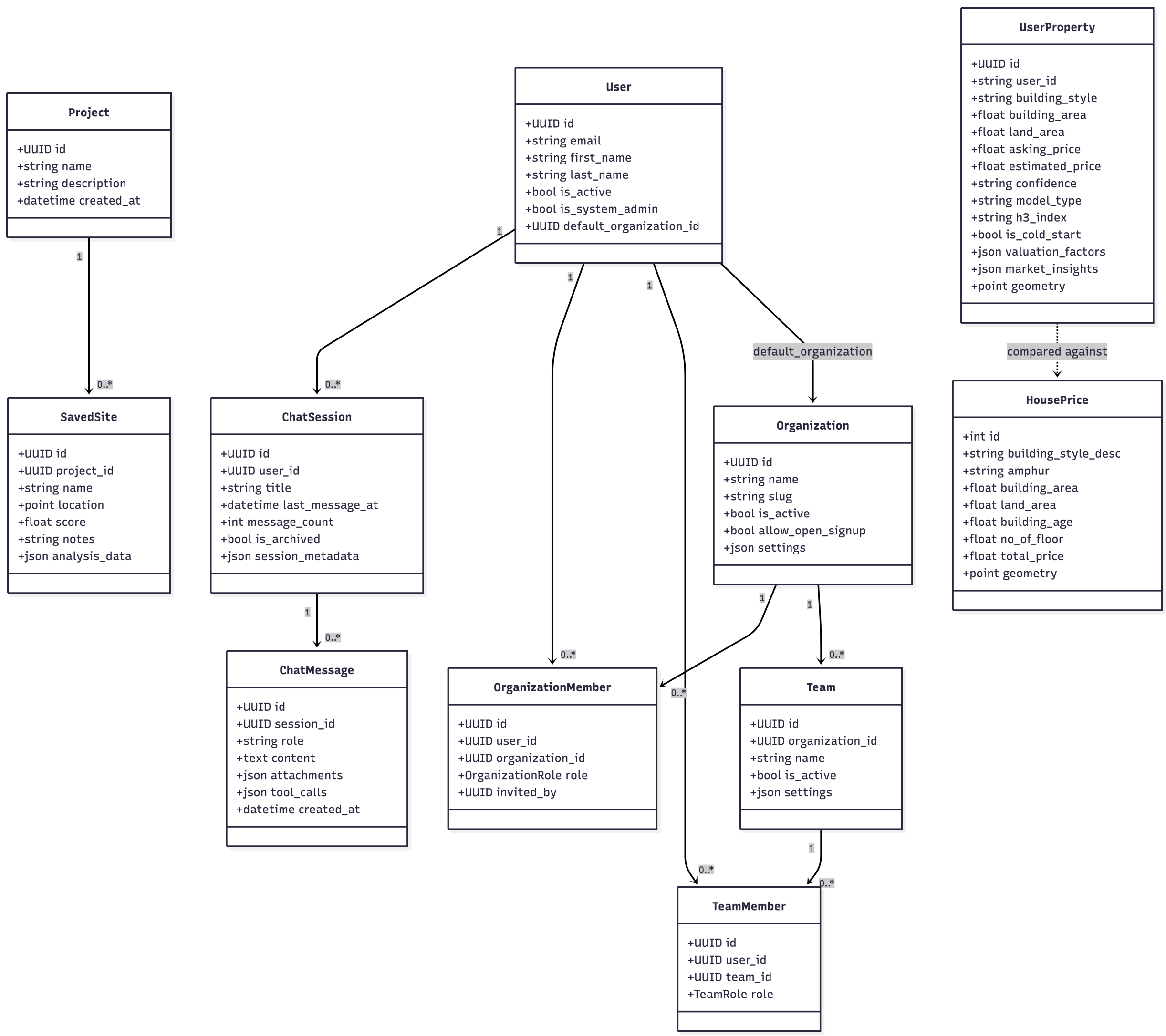


Figure 4.1: Domain Model of the BorBann Platform

Figure 4.2 presents the design class diagram.

The diagram reflects a service-oriented internal structure. Route handlers receive requests from the frontend and delegate work to specialized service classes such as the price-prediction service, location-intelligence service, catchment service, and agent-related orchestration services. Model classes define the data exchanged across these layers, while the persistence layer manages database interaction and storage. This separation improves maintainability and allows predictive components, geospatial analytics, and user-facing APIs to evolve without tightly coupling all parts of the system.

4.3 Sequence Diagram

The sequence diagram describes the order of interaction among the main system components during a representative user workflow. For BorBann, an important scenario is

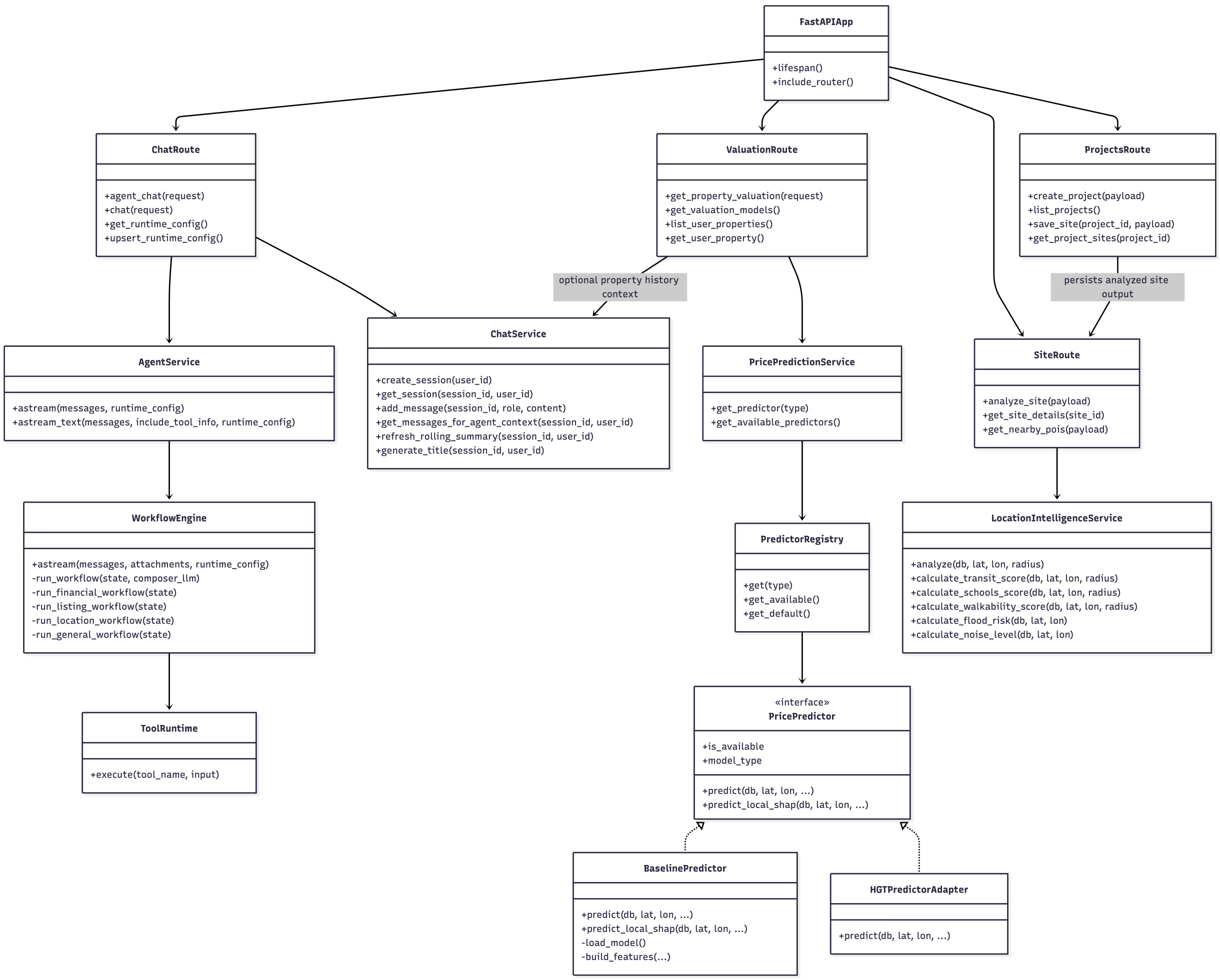


Figure 4.2: Design Class Diagram for BorBann

the user request for property-related analysis through the frontend, followed by backend processing, service invocation, data retrieval, and response generation.

Figure 4.3 shows the sequence diagram for the system.

This interaction begins when the user initiates an action from the frontend interface, such as a valuation request, property lookup, or agent-assisted question. The request is transmitted to the backend API, which validates input and delegates the task to the relevant service layer. The selected service then communicates with the database, model runtime, or geospatial logic as needed. Once the required outputs are generated, the response is returned to the frontend and rendered as property details, valuation results, location scores, or conversational feedback. The sequence diagram highlights the coordinated processing required to combine multiple subsystems into one coherent user interaction.

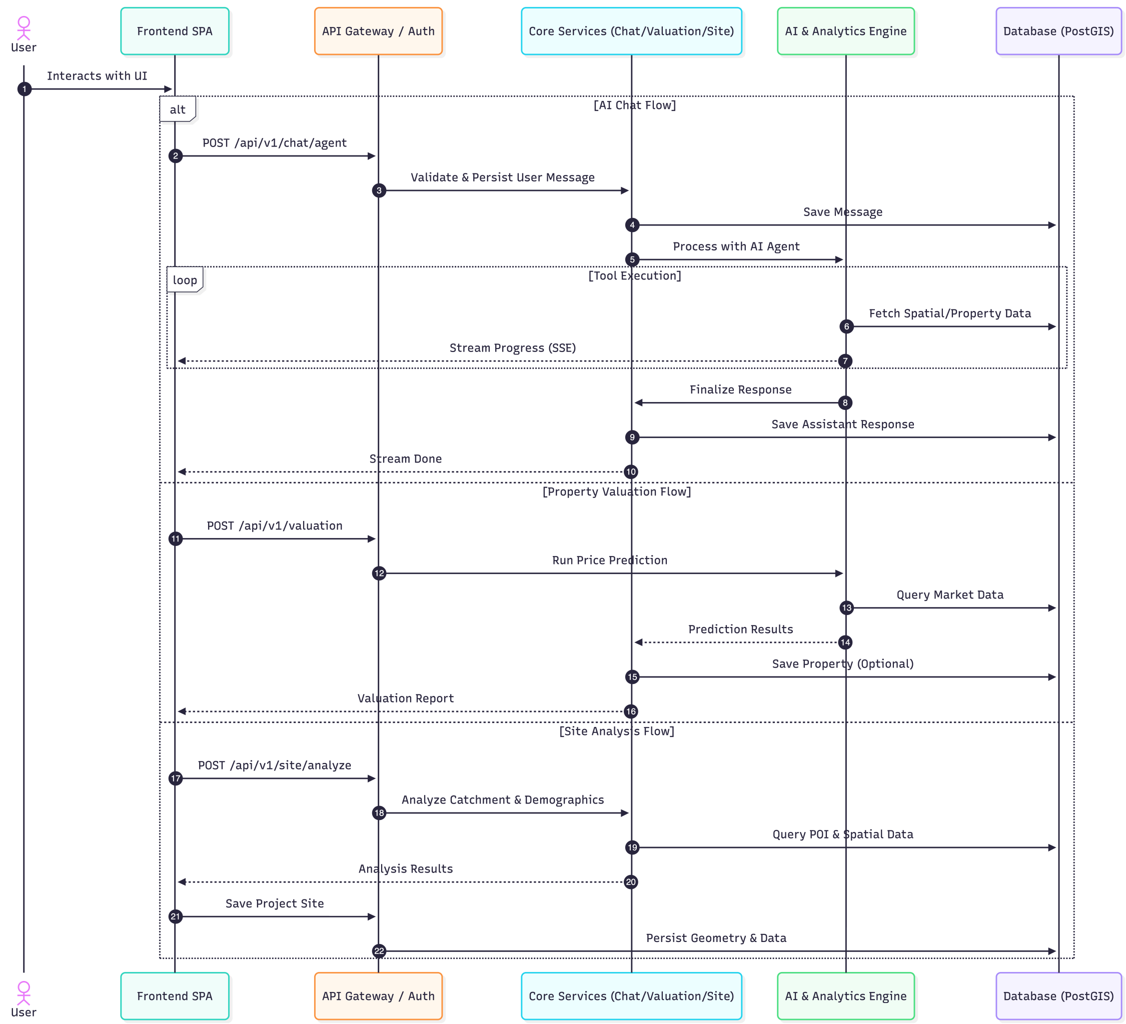


Figure 4.3: Sequence Diagram of a Typical BorBann Interaction

4.4 System Architecture

The system architecture provides the highest-level technical view of BorBann. It describes how the frontend, backend, database, AI services, and supporting infrastructure operate together as a complete platform. This view is important because BorBann is not a single-model application; instead, it integrates user interaction, geospatial processing, valuation logic, and explainability support within one multi-layer software system.

Figure 4.4 presents the system architecture of the platform.

At the presentation layer, the frontend provides map-based exploration, property viewing, valuation workflows, and conversational interaction. The application layer is implemented through FastAPI services that expose endpoints for authentication, property access, chat, valuation, price prediction, and geospatial analytics. These services coordinate with model-oriented and geospatial service layers, which perform prediction, explanation handling, catchment analysis, and location scoring. At the data layer, PostgreSQL with

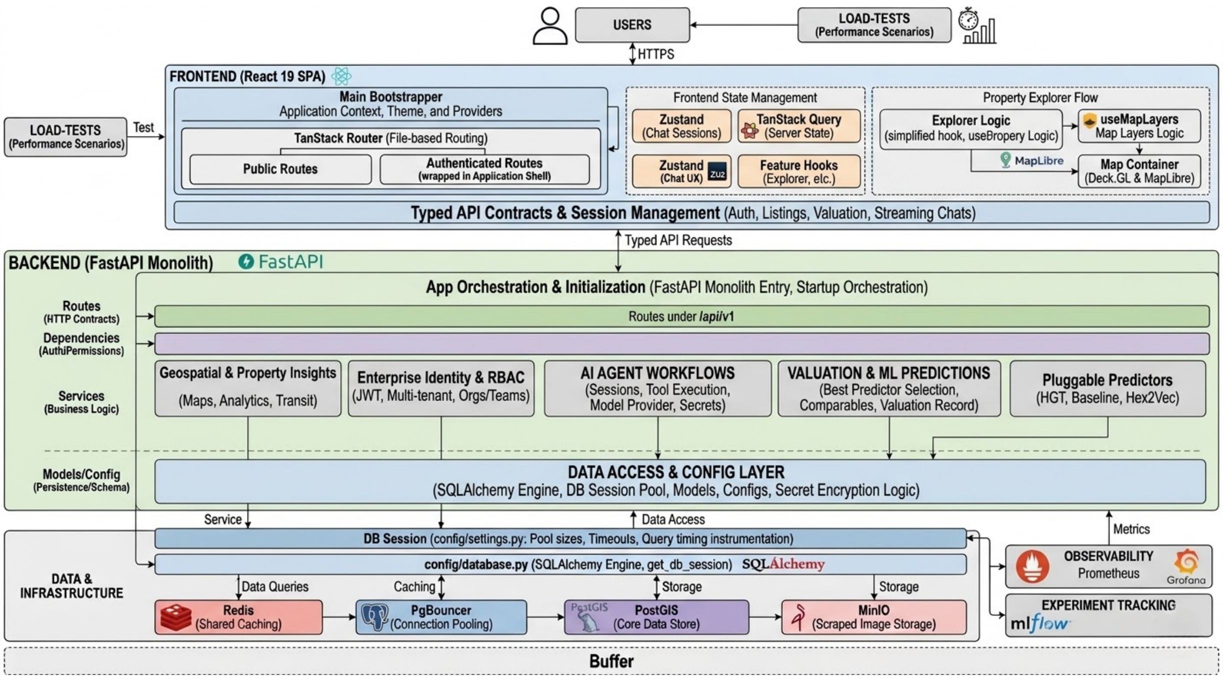


Figure 4.4: System Architecture of BorBann

PostGIS stores spatially enabled real-estate and point-of-interest data, while related run-time assets and models support valuation and AI-driven responses. This layered architecture enables BorBann to integrate interactive web functionality with AI-enabled decision support in a structured and extensible manner.

Chapter 5

AI Component Design

This chapter describes the AI-related architecture implemented in `site-selection-core`. In the current system, the main AI functionality is organized around a workflow-driven conversational agent, deterministic location intelligence, and house price prediction with valuation-oriented explanation. Commercial site analysis remains available as a supporting geospatial utility, but the principal AI focus of the platform is residential property price estimation and interpretation.

5.1 Business Context and AI Integration

Figure 5.1 shows the overall workflow of the platform. In the current implementation, AI is not introduced as a single isolated model. Instead, it is integrated across three tightly connected layers: conversational orchestration, geospatial scoring, and house price prediction. This design reflects the practical nature of real estate decision support, where the user usually needs a combination of search, explanation, neighborhood context, and price estimation rather than one standalone prediction.

Table 5.1 summarizes the AI-related subsystems that are actually implemented in the repository.

The core AI problem addressed by BorBann is house price prediction and explanation, supported by a conversational interface and location-based analytics.

5.2 Goal Hierarchy

The goal hierarchy of the AI component can be understood from both strategic and operational perspectives.

Principal AI Components

The AI component of BorBann is organized around four principal implementation layers.

First, the conversational agent is implemented as a workflow-driven orchestration service. Instead of relying on unconstrained response generation, the backend normalizes

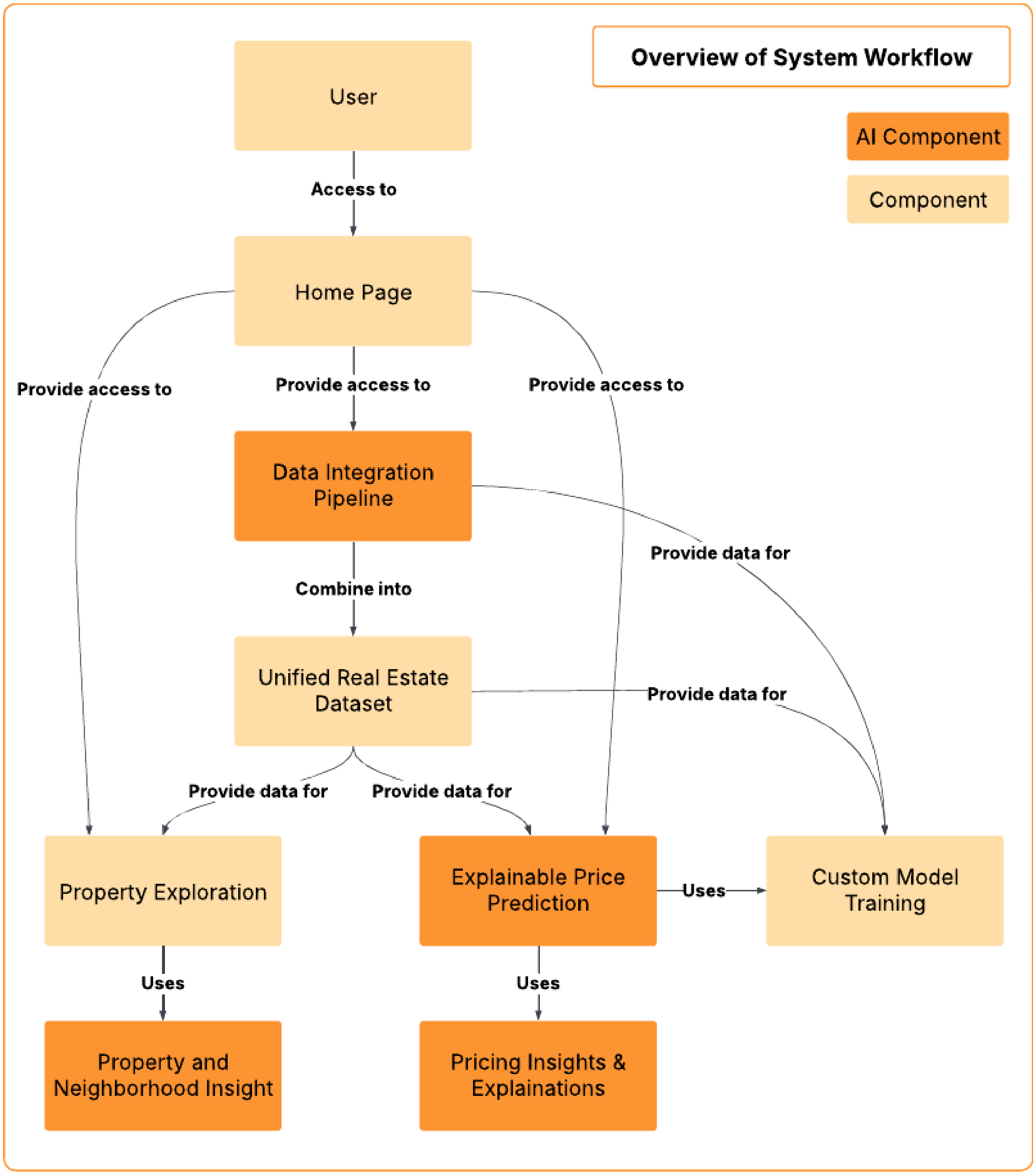


Figure 5.1: Overall System Workflow

Subsystem	Implementation Form	Current Role in the Platform
Conversational agent	Workflow-based backend service with streaming chat API	Interprets user requests, loads session context, routes to the appropriate workflow, executes tools, and returns evidence-backed responses.
Location intelligence engine	Deterministic geospatial service	Produces composite location scores and interpretable sub-scores for transit, walkability, schools, flood risk, and noise exposure.
House price prediction engine	Pluggable prediction service with multiple model adapters	Generates property price estimates, confidence signals, district comparisons, feature contributions, and valuation-oriented output for the user interface.
Explainability layer	Runtime explanation formatting plus offline evidence loading	Exposes local factor explanations and benchmark-oriented explainability evidence for transparency and accountability.
Supporting site analytics	Rule-based geospatial utility and site-inspector page	Provides competitor, magnet, and catchment summaries, but acts as a secondary analytic utility rather than the principal AI component of the system.

Table 5.1: Current AI-Related Subsystems in the Implementation

requests, routes them to workflows, executes tools, verifies results, and streams structured progress events such as `thinking`, `step`, `token`, `error`, and `done`. This design supports trustworthy interaction in a domain where many user queries depend on factual claims about prices, districts, and locations.

Second, the location intelligence engine provides deterministic geospatial scoring. It computes interpretable indicators for transit accessibility, walkability, school access, flood risk, and noise exposure. Because these outputs are rule-based and spatially grounded, they can be validated and reused across property detail, map exploration, and valuation workflows.

Third, the main predictive AI service is house price prediction. The repository implements a pluggable prediction architecture with a shared interface and lazy-loading model registry. The service supports baseline LightGBM prediction, Hex2Vec-enhanced prediction, and HGT-based prediction when the required model assets are available.

Fourth, the valuation layer transforms prediction output into an end-user report. Instead of returning only one numeric estimate, the service also provides price range, confidence category, price per square meter, factor-level explanation, comparable nearby properties, district-level market context, and optional valuation-history persistence.

Within this architecture, commercial site analysis remains a supporting geospatial utility. It contributes secondary neighborhood and catchment-oriented inspection, but the principal AI contribution of the platform remains residential property price prediction and interpretation.

Organizational Goals

- To establish a trustworthy real estate information platform that integrates geospatial evidence, prediction, and explanation in one system.
- To demonstrate that AI can support practical property decision making when combined with interpretable outputs and operational controls.

System Goals

- To provide reliable AI-assisted workflows across chat, location intelligence, and valuation services.
- To maintain a modular service architecture so that predictive backends can evolve without changing the external platform contract.
- To support production-oriented deployment, observability, and operational governance for AI-enabled services.

User Goals

- To allow users to ask property-related questions in natural language and receive grounded answers.
- To help users understand neighborhood context through interpretable spatial indicators.
- To provide understandable reasons behind estimated property values instead of only a single opaque output.

AI Model Goals

- **Conversational agent:** route requests correctly, preserve session context, and use tools before making factual claims.
- **Location intelligence:** produce stable and interpretable composite and sub-scores from explicit geospatial rules.
- **House price prediction:** estimate residential property value from structural and spatial inputs while supporting multiple prediction backends.

- **Valuation layer:** convert model output into a report suitable for end users through confidence signals, factor summaries, comparable properties, and district-level context.

Three predictor types are represented in the repository:

Predictor Type	Repository Representation	Role
Baseline	LightGBM-based baseline predictor	Provides the standard tabular price prediction path using spatial and structural property features.
Baseline + Hex2Vec	LightGBM predictor with Hex2Vec features	Extends the baseline with spatial embeddings to capture neighborhood context more richly.
HGT	Heterogeneous Graph Transformer adapter	Supports graph-based experimentation and can be selected when the required model assets are available.

Table 5.2: Prediction Backends Exposed by the Current Price-Prediction Service

The runtime registry selects the best available predictor automatically according to model availability. In the present implementation, the default priority is HGT first, followed by the Hex2Vec-enhanced baseline, and then the plain baseline. This design supports experimentation without rewriting the service interface.

5.3 AI Task Requirements Analysis Using AI Canvas

This section presents the main AI tasks using the AI Canvas structure of Requirement, Specification, and Environment. In this chapter, the AI Canvas is described in narrative form rather than as a diagram because the current objective is to clarify the operational role of each AI component in the system.

Conversational Agent

Requirement: The conversational agent must interpret natural-language property questions, preserve session continuity, use tools before returning factual responses, and stream execution progress to the user interface.

Specification: The implementation satisfies this requirement through workflow-based orchestration with request normalization, routing, typed tool execution, response composition, verification, session persistence, rolling summaries, and server-sent event streaming.

Environment: The agent operates in an interactive multi-turn web environment with optional spatial attachments, authenticated sessions, model-provider variability, and the need to produce evidence-backed responses under practical latency constraints.

Location Intelligence Engine

Requirement: The location intelligence engine must produce stable, interpretable, and reusable indicators for accessibility, amenities, schools, flood risk, and environmental exposure.

Specification: This requirement is addressed through deterministic geospatial scoring rules that compute transit, walkability, schools, flood, and noise outputs and combine them into a composite location score on a 0–100 scale.

Environment: The component operates over spatial database queries involving Bangkok property data, points of interest, transit infrastructure, and environmental-risk datasets, and it is reused across property pages, valuation workflows, and map-based exploration.

House Price Prediction and Valuation

Requirement: The house price prediction component must estimate residential property prices from structural and spatial inputs while also supporting explanation-oriented outputs suitable for user-facing valuation reports.

Specification: The implementation uses a pluggable predictor interface that supports baseline LightGBM, Hex2Vec-enhanced baseline, and HGT-based prediction paths. On top of this prediction layer, the valuation service provides price range, confidence category, comparable properties, district insights, and formatted factor descriptions.

Environment: This component operates in a mixed real-estate data environment with spatial attributes, incomplete or evolving model artifacts, district-level variation, and a product context that requires both prediction quality and interpretability.

5.4 User Experience Design with AI

The AI-related user experience is designed around clarity, guidance, and evidence presentation. The user is not expected to understand internal model architecture. Instead, the interface translates technical processing into practical outputs that support property exploration and decision making.

Table 5.3 summarizes the main AI-supported user interactions.

This means that AI in BorBann is not presented as a separate experimental feature. It is embedded into normal user tasks such as asking questions, reviewing a property, understanding neighborhood conditions, and estimating value.

5.5 Deployment Strategy

The production deployment architecture for the current implementation is based on a single-VPS model using Docker Compose, host-level Nginx, Tailscale-restricted administration, and GitHub Actions deployments through GitHub Container Registry (GHCR).

User Interaction	AI or Analytic Support	Visible User Outcome
Conversational property inquiry	Workflow-based chat agent with tool-backed reasoning	The user can ask questions in natural language and receive structured answers grounded in retrieved evidence and platform data.
Property valuation request	Price-prediction and valuation service	The user receives an estimated price, confidence level, explanatory factors, nearby comparables, and district market context.
Property-level explanation review	Runtime explanation formatting and explainability evidence	The interface presents factor-level signals and explanation-oriented narrative rather than only one opaque number.
Neighborhood review	Location intelligence engine	The user sees interpretable sub-scores for accessibility and risk-related context around a property.
Secondary site inspection	Rule-based site analytics and isochrone support	The interface provides supporting geospatial context, but this remains separate from the central house-price AI workflow.

Table 5.3: Main User Experience Flows Supported by AI or Analytics

Table 5.4 summarizes the current deployment architecture.

This deployment design has several implications for the AI component. First, the AI services are packaged as part of the backend application rather than being deployed as isolated model-serving products. Second, heavy runtime assets such as trained models and geospatial data are mounted into the deployment environment, which reduces rebuild cost and separates code rollout from data synchronization. Third, deployment safety is improved through automated testing, image tagging, health checks, and rollback scripts.

The CI/CD workflow runs backend and frontend validation, builds backend, frontend, and database images, pushes them to GHCR, connects to the production VPS through Tailscale, performs deployment by script, and then verifies service health. This deployment structure supports the operational requirements of the AI-enabled platform while preserving a practical and maintainable release process.

Layer	Current Implementation	Operational Purpose
Public ingress	Host Nginx on ports 80 and 443	Terminates public traffic and forwards requests to loopback-bound frontend and backend services.
Frontend service	Containerized frontend image	Serves the web interface and exposes only a loopback-bound host port in production.
Backend service	Containerized FastAPI image	Hosts chat, valuation, price prediction, location intelligence, authentication, and analytics endpoints.
Database layer	PostGIS database plus PgBouncer	Supports geospatial queries and connection pooling for application traffic.
Cache and object storage	Redis and MinIO	Provide shared caching and image or object storage support for runtime features.
Runtime data mounting	Mounted runtime data and model directories	Keeps large datasets and trained model assets outside the application image for smaller, code-focused deployments.
Private administration	Tailscale-restricted access	Limits SSH, database administration, and optional observability access to the private mesh network.
Deployment automation	GitHub Actions plus GHCR	Runs tests, builds images, pushes tagged artifacts, deploys over Tailscale SSH, performs health checks, and supports rollback.

Table 5.4: Current Production Deployment Architecture

Chapter 6

Software Development

This chapter provides an overview of the software development process for the **Borbann** platform, detailing the key practices, technologies, and tools used to build and maintain the application. Effective software development is essential for creating a high-quality product that meets user needs and aligns with business goals. In this chapter, we discuss the chosen technology stack, the established coding standards, and the methodologies used throughout the development lifecycle.

6.1 Software Development Methodology

Chosen Methodology

For the development of Borbann, we adopted an **Iterative and Incremental** development methodology. This approach allows the team to break down the complex requirements of a real estate AI platform into manageable components, delivering value in stages.

Key reasons for selecting this methodology include:

- **Flexibility:** Allows adjustments based on model performance results, such as switching from baseline LightGBM to Graph Neural Networks.
- **Continuous Integration:** Facilitated by tools like Dagger and Docker, ensuring that code changes are regularly tested and integrated.
- **Parallel Development:** Enables the frontend and backend teams to work simultaneously on defined interfaces (APIs).

Work Process

The work process follows a systematic approach to ensure efficiency and scalability, divided into key stages:

1. **Requirement Analysis & Feature Definition**

- Analyzing the needs for property valuation and location intelligence.
- Defining core features such as the AI Valuation Model, Chat Agent, and Interactive Map.

2. **System Design & Architecture**
   - **Database Schema:** Designing geospatial tables using PostGIS to handle complex location data effectively.
   - **API Contract:** Defining RESTful endpoints using FastAPI's Pydantic models to ensure type safety between frontend and backend.

3. **Development Phase**
   - **Backend & AI:** Developing the GIS server, training production ML models (LightGBM + Hex2Vec), and exposing them via APIs.
   - **Frontend:** Building the interactive user interface with React and integrating map visualizations using Deck.gl and MapLibre.

4. **Testing & CI/CD**
   - Automated testing using `pytest` for backend and `vitest` for frontend.
   - Continuous integration pipelines managed by Dagger to ensure build stability.

6.2 System Architecture in Development Context

The software development process of BorBann is closely tied to its system architecture. The project is not implemented as a single monolithic application; rather, it is developed as an integrated platform in which frontend interaction, backend APIs, AI services, geospatial analytics, and database infrastructure evolve together. This architectural perspective is important during development because each implementation task must preserve the interfaces between these layers.

Figure 6.1 shows the system architecture referenced during development.

From a development perspective, the frontend layer is responsible for interactive map rendering, property views, valuation workflows, and conversational user interfaces. The backend layer provides FastAPI-based endpoints that mediate access to prediction, location intelligence, search, authentication, and chat services. The service layer contains the AI-oriented and geospatial logic used to support valuation, explainability, catchment analysis, and neighborhood scoring. Finally, the database and runtime asset layer store geospatial records, property data, and trained-model artifacts required by the application.

This architecture influenced implementation planning in several ways. It enabled the frontend and backend to progress in parallel through clearly defined API contracts. It also encouraged service modularity so that price-prediction logic, explainability support, and geospatial processing could be improved without destabilizing unrelated parts of the platform. For this reason, the architecture served not only as a design artifact but also as a practical guide for incremental development.

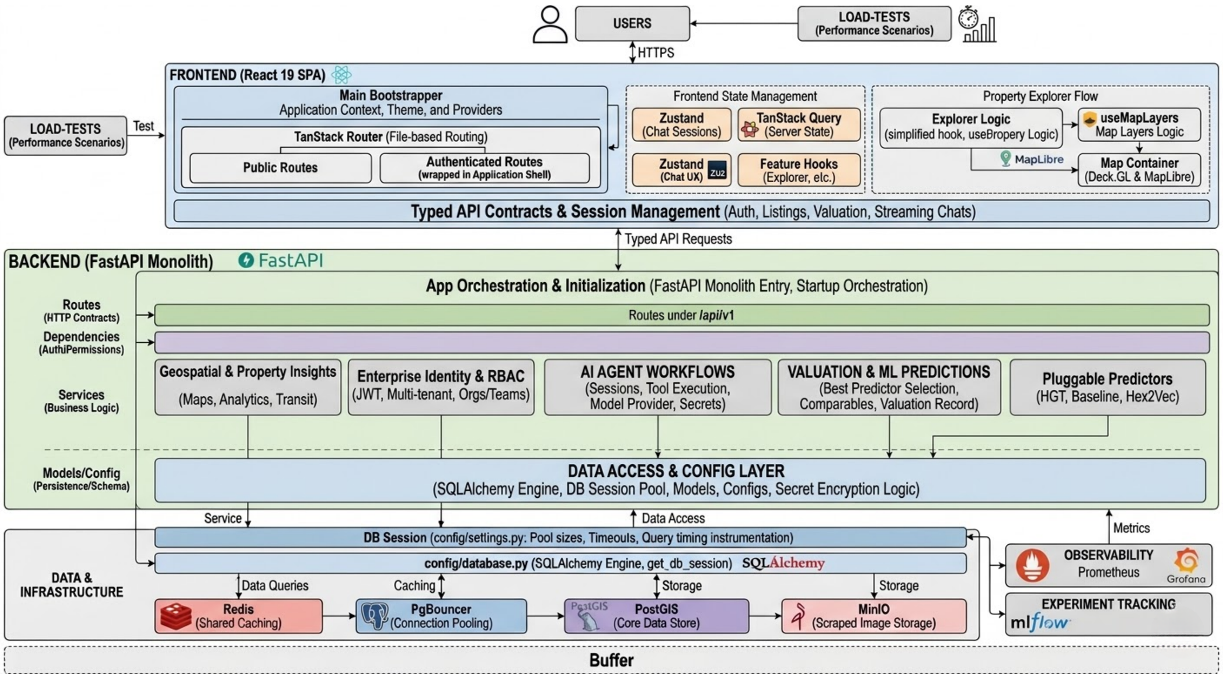


Figure 6.1: System Architecture Referenced During Software Development

6.3 Technology Stack

To develop the Borbann platform, we utilized a modern, performance-oriented technology stack optimized for geospatial data and AI workloads.

Frontend Development

- **React (v19):** A JavaScript library for building user interfaces, utilized with TypeScript for type safety.
- **Vite:** A build tool that provides a fast development environment and optimized production builds.
- **Tailwind CSS (v4):** A utility-first CSS framework for rapid and responsive UI design.
- **TanStack Query & Router:** For efficient server state management and client-side routing.
- **Deck.gl & MapLibre:** High-performance WebGL-powered libraries for rendering large-scale geospatial data and interactive maps.
- **Radix UI:** A library of unstyled, accessible components for building high-quality design systems.

Backend Development

- **Python (3.11+):** The primary programming language, chosen for its strong support in data science and web frameworks.
- **FastAPI:** A modern, high-performance web framework for building APIs with Python and asynchronous programming support.
- **SQLAlchemy & GeoAlchemy2:** ORM tools for interacting with the database and handling spatial data types.
- **UV:** A fast Python package installer and resolver significantly improving dependency management speed.

Database Management

- **PostgreSQL:** A powerful, open-source object-relational database system.
- **PostGIS:** A spatial database extender for PostgreSQL, essential for storing and querying location data (e.g., geometries, distances).

AI & Machine Learning

- **PyTorch & PyG (Torch Geometric):** Used for developing and training Graph Neural Networks (Heterogeneous Graph Transformer).
- **LightGBM:** A gradient boosting framework used for baseline property price prediction models.
- **LangChain & LangGraph:** Frameworks for building the AI Chat Agent and orchestrating LLM interactions.
- **MLflow:** Used for experiment tracking, model visioning, and managing the ML lifecycle.

DevOps & Tools

- **Docker:** Containerization platform ensuring consistent environments across development and production.
- **Dagger:** A programmable CI/CD engine used to define and run delivery pipelines.
- **Biome:** A fast formatter and linter for the frontend codebase.
- **Make:** Automation tool used to simplify common tasks like starting the stack, running tests, and managing database migrations.

6.4 Coding Standards

Coding standards are enforced to ensure maintainability, readability, and consistency across the codebase.

General Practices

- **Environment Security:** Sensitive configuration and credentials are managed via `.env` files and never committed to version control.
- **Git LFS:** Large files, particularly ML models and datasets, are managed using Git Large File Storage.

Backend Standards (Python)

- **Type Hinting:** Comprehensive use of Python type hints to improve code clarity and enable static analysis.
- **Path Handling:** Usage of `pathlib` over `os.path` for robust, cross-platform file system operations.
- **Modular Structure:** routes, models, and business logic are separated into distinct directories (`src/routes`, `src/models`, `src/services`).
- **Package Management:** Strict usage of `uv` for adding and managing dependencies to ensure reproducible builds compared to standard pip.

Frontend Standards (TypeScript/React)

- **Accessibility (a11y):** Strict adherence to WAI-ARIA guidelines (e.g., valid `role` attributes, meaningful `alt` text, keyboard navigability).
- **Linting & Formatting:** `Biome` is used to enforce code style and catch errors automatically.
- **Component Structure:** Functional components using Hooks, with a focus on reusability and separation of concerns.
- **Performance:** Utilization of `React.memo` and efficient state management to prevent unnecessary re-renders, especially for map components.

6.5 Progress Tracking

Progress throughout the development lifecycle was tracked using a combination of version control analysis and task management.

6.4.1 Development Velocity

The project followed a phased implementation strategy:

1. **Phase 1: Foundation.** Setup of the PostGIS database, Dagger pipelines, and basic FastAPI structure.
2. **Phase 2: Core ML.** Data processing pipelines, training of the LightGBM baseline, and initial HGT (Heterogeneous Graph Transformer) experiments.
3. **Phase 3: Integration.** Development of the React frontend, integration of Mapbox/Deck.gl for visualization, and connection to backend APIs.
4. **Phase 4: Refinement.** UI/UX improvements, implementation of the AI Chat Agent using LangGraph, and performance optimization.

Regular commits and pull requests were used to maintain a steady stream of updates, with the ‘Makefile‘ serving as the central hub for running development tasks and continuous integration checks.

6.4.2 Rebuttal to Committee Review

After the committee review, the team converted each major concern into an implementation task and validated the result through code changes, evaluation scripts, and interface updates. The goal of this revision cycle was not only to answer comments in writing, but to improve the system in a measurable way.

In conclusion, the committee feedback led to practical improvements in explainability, transparency, interface readability, and chatbot grounding. Some items, especially full source-link completion and further chatbot quality recovery, still require continued refinement. However, the project now has clearer evidence, better user-facing guidance, and stronger validation workflows than in the earlier review stage.

Issue Raised	Resolution in BorBann	Current Outcome
Proof and evaluation of price factors	Added stronger explainability support through per-request local SHAP factors, district and H3 comparison metrics, and an explainability evaluation pipeline that measures rank correlation, top-k overlap, perturbation faithfulness, and expected-feature coverage.	The valuation module now provides measurable evidence that major factors affect the prediction, instead of showing unsupported descriptive factors only.
Tooltip and data sources	Added reusable source tooltips and a source registry for key indicators such as rail transit, flood risk, POI-based signals, and valuation-related data.	Users can inspect the origin of major indicators directly from the interface, while the remaining placeholder links are tracked for completion.
UI colors and legend clarity	Added explicit score legends for location intelligence, trend legends for market insight cards, and clearer H3 legend support for map-based layers.	Visual components are easier to interpret and no longer rely only on unstated color meaning.
Protection against model hallucination	Kept the stable LightGBM plus Hex2Vec path as the production baseline, compared candidate models against benchmark acceptance gates, and added faithfulness checks and baseline comparisons for explanation quality.	We avoid promoting higher-risk models that do not outperform the baseline, and we provide a stronger basis for trusting the explanation output.
Chatbot retrieval accuracy	Improved chatbot grounding with spatial context injection, internal neighborhood knowledge, pgvector-based retrieval, referenced responses, and a golden-set quality benchmark.	Answer quality for location-specific questions is better grounded in evidence, and remaining misses are now visible through repeatable evaluation instead of anecdotal testing.

Table 6.1: Summary of Issues and Resolutions in BorBann

Chapter 7

Deliverables

This chapter presents the main deliverables produced by the BorBann project. Since BorBann is a multi-page platform rather than a single demo screen, the delivered outputs include several user-facing workflows, backend and AI services, evaluation assets, and deployment materials. Each deliverable is intended to support a specific part of the real estate analysis process, from account access and property exploration to valuation, site analysis, and operational monitoring.

The implementation encompasses 11 user interface routes, comprising 2 public and 9 authenticated access points. The system architecture includes 15 backend testing modules, a continuous integration pipeline, and a comprehensive data foundation processing over 800,000 spatial records (points of interest, transit stops, real estate records, and demographic grid cells). This demonstrates a fully functional, production-ready system.

7.1 Overview of Delivered Outputs

Table 7.1 groups the delivered outputs by their practical role in the platform. It shows that the final scope includes both user-visible workflows and the supporting technical assets required to keep those workflows reliable.

7.2 Detailed User-Facing Deliverables

BorBann delivers multiple pages and workflows to support different user tasks. Instead of concentrating all functionality on one map page, the platform separates each major task into a dedicated screen so that the user can move from exploration to evaluation and then to deeper analysis.

Table 7.2 summarizes the 11 end-user route files delivered in the frontend. The table focuses on the main workflow each page supports and the concrete capability exposed to the user.

The delivered pages work together as a complete workflow. A user can enter the platform through authentication, explore the map, open a property or listing detail page,

Deliverable Group	Main Form	Scope of Delivery
User access module	Public pages	Login and registration workflows for controlled platform access.
Property exploration module	Main authenticated pages	Interactive map, filters, legends, overlays, property popups, and district-based browsing.
Property detail module	Detail pages	Detailed property and listing pages with map context, nearby properties, and location intelligence.
Valuation module	Wizard and report pages	Input form, location picker, AI valuation report, explainable factors, comparables, and downloadable report.
Chat and guided analysis module	Session pages	AI chat interface with session flow, map context, references, and tool-assisted answers.
Site analysis module	Inspector page	Site score, magnets, competitors, isochrone visualization, and exportable site report.
Market analytics module	Analytics page	District-level summary statistics and sortable market comparisons.
System configuration module	Settings page	Runtime provider configuration, validation, and account management controls.
Data and AI backend	Services and APIs	FastAPI APIs, PostGIS queries, retrieval services, and model inference endpoints.
Quality and deployment package	Reports and runbooks	Automated tests, benchmark reports, load tests, CI workflows, and production deployment assets.

Table 7.1: Overview of BorBann Deliverables

request a valuation, inspect the surrounding area, ask the chatbot for clarification, and continue to district or site-level analysis when needed.

Figures 7.1 to 7.9 show the implemented user interfaces that correspond to the delivered workflows listed in Table 7.2.

Page or Workflow	Main Purpose	Delivered Capability
Login page	User authentication	Secure sign-in flow for returning users with access to the authenticated platform.
Registration page	Account creation	New-user registration with profile information and password-strength guidance.
Main property explorer	General property discovery	Full-screen map explorer with filters, H3 heatmap layers, overlay controls, legends, source references, and property count summary.
Property popup flow	Quick map inspection	Click-based property preview from the map before opening a full detail page.
Property detail page	Official property record review	Detailed property information, nearby property comparison, embedded location intelligence, and comprehensive price report view.
Listing detail page	Listing-centric review	Listing information with images, source label, nearby property context, location intelligence, and comprehensive price reporting.
Valuation input page	Valuation request creation	Property upload form and map-based location picker for requesting a valuation.
Valuation report page	Explainable valuation output	Estimated price, confidence signal, factor breakdown, comparables, market insights, downloadable PDF, sharing support, and linked location intelligence.
District analytics page	Area-level market understanding	District-by-district property counts, average prices, price-per-square-meter indicators, sorting, and search support.
AI chat welcome and session pages	Conversational analysis	Session-based AI interaction for property search, location questions, and guided reasoning with map context and references.
Site inspector page	Business site evaluation	Site score, 15-minute isochrone, nearby competitors, nearby magnets, traffic potential view, and report export.
Settings page	Runtime and account management	Provider selection, runtime validation, secure configuration storage, and account controls.

Table 7.2: Detailed User-Facing Deliverables

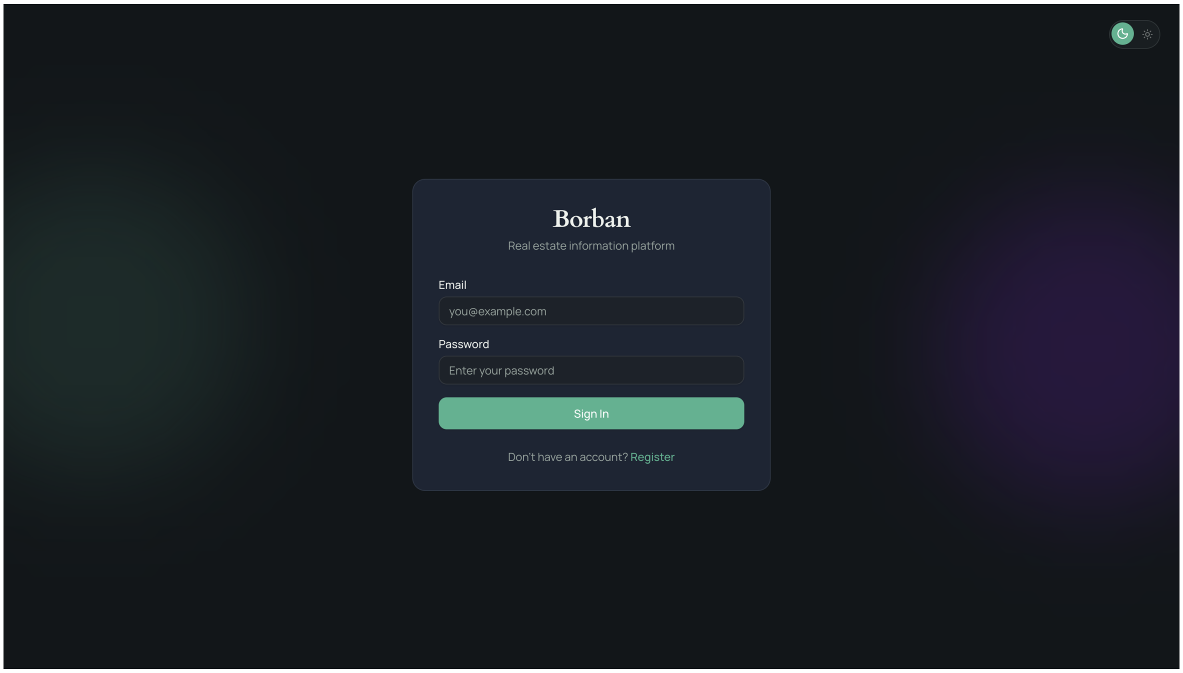


Figure 7.1: BorBann Login Page

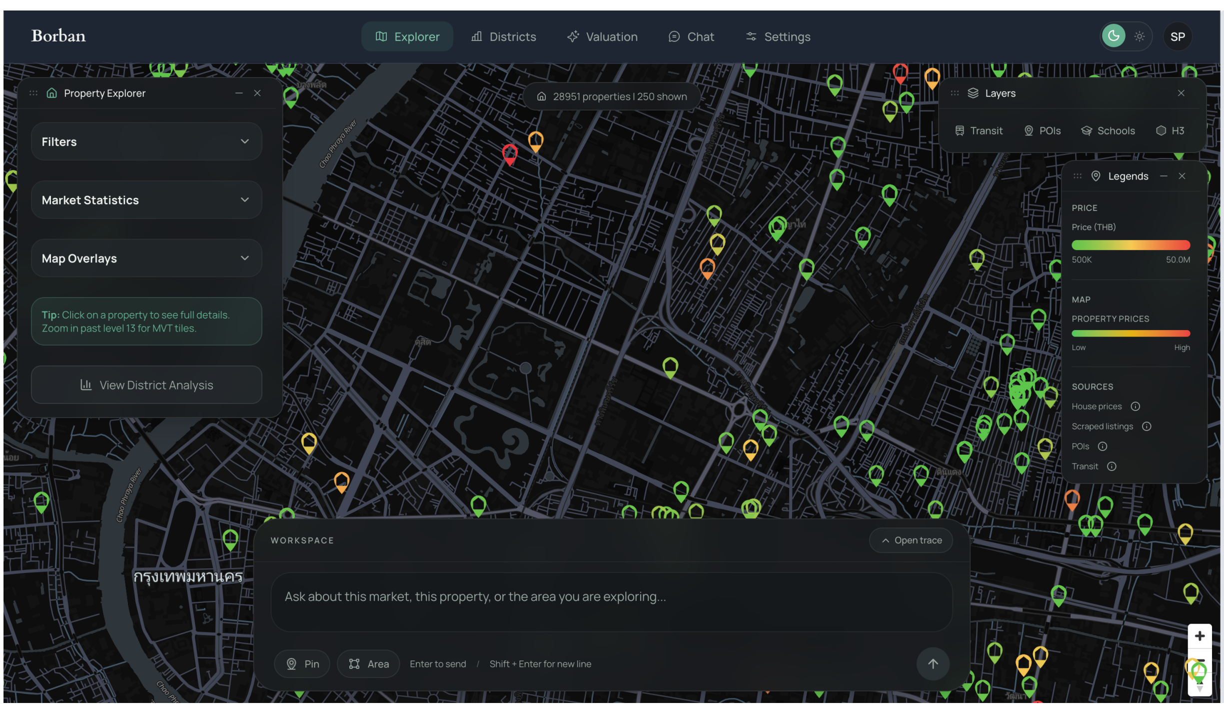


Figure 7.2: Main Property Explorer Interface

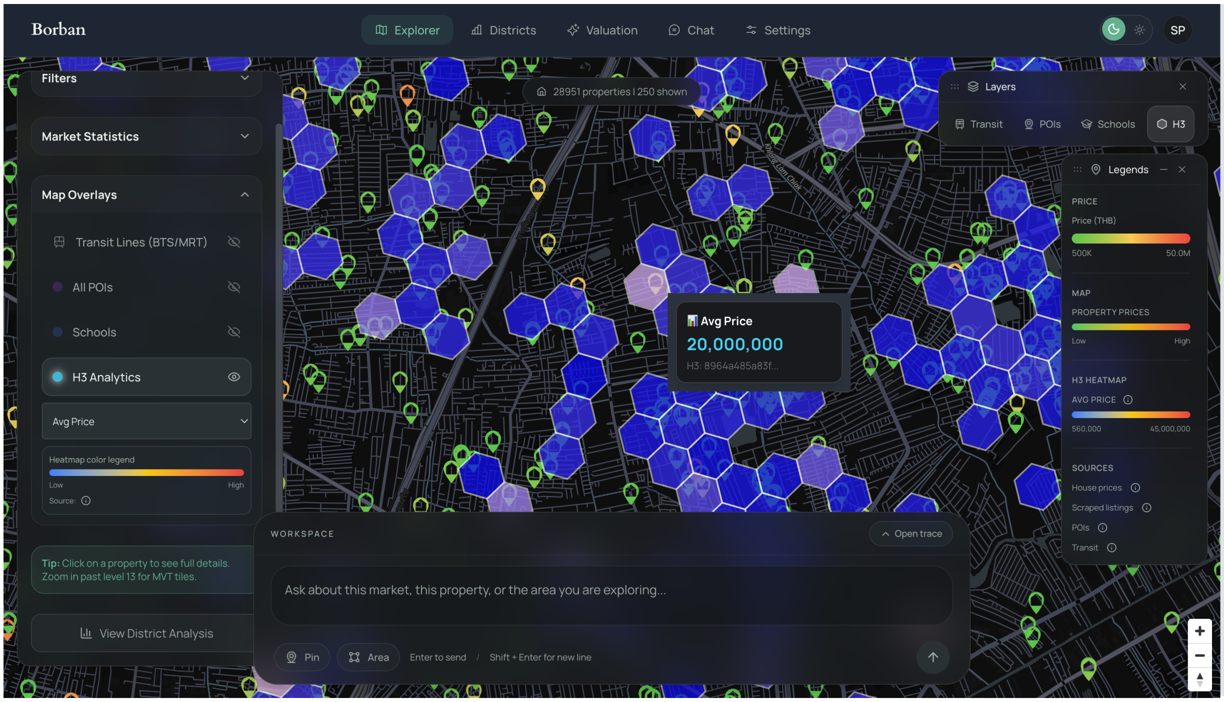


Figure 7.3: Explorer Interface with H3 Analytics Layer

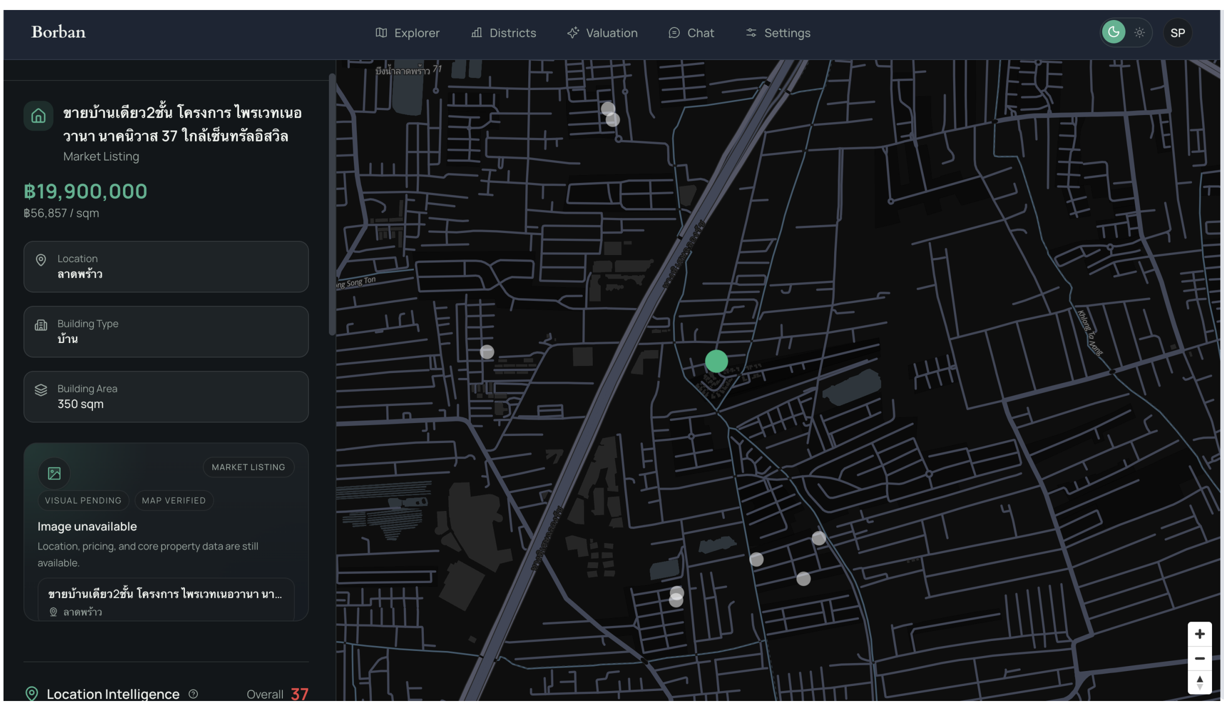
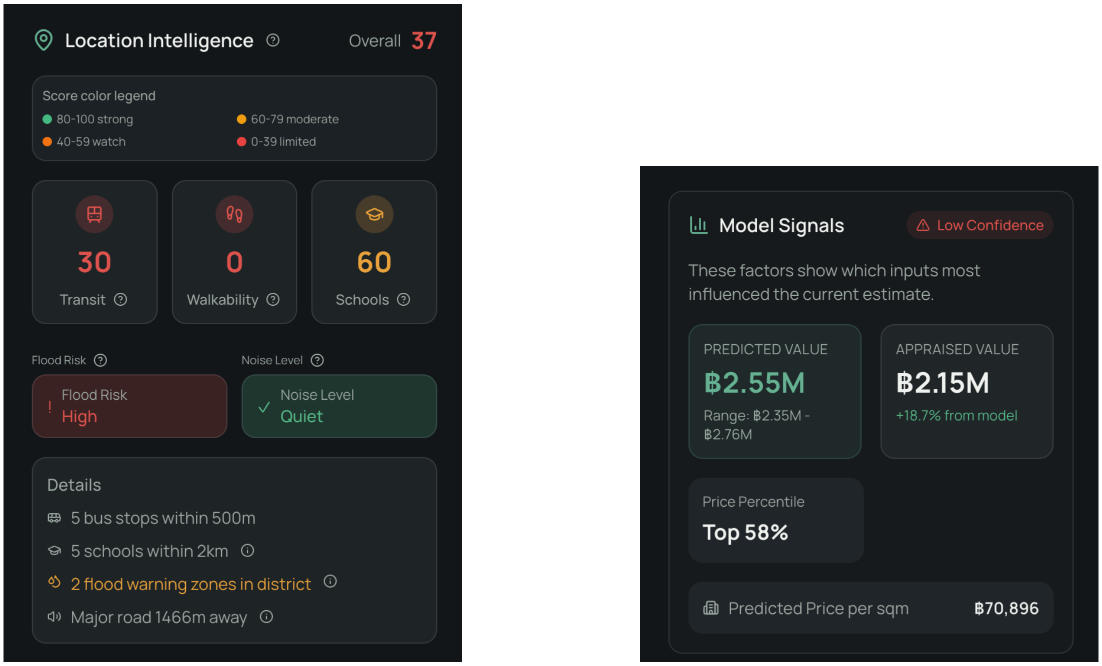


Figure 7.4: Property Detail Page



(a) Location intelligence panel (b) Model signal summary

Figure 7.5: Property-Level Analytical Panels

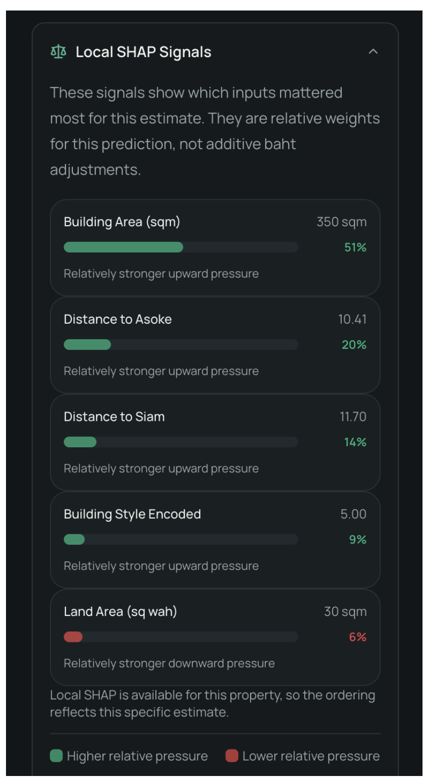
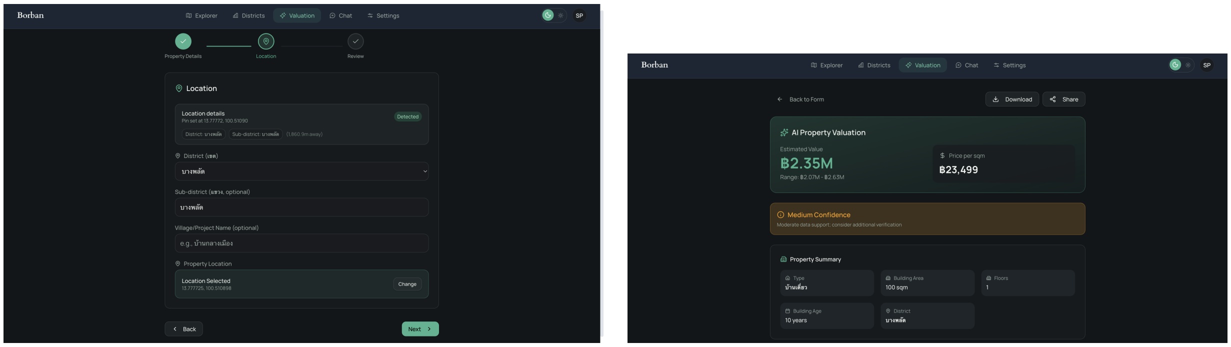


Figure 7.6: Property Page Local SHAP Explanation



(a) Valuation request form (b) Valuation report page

Figure 7.7: Valuation Workflow Interfaces

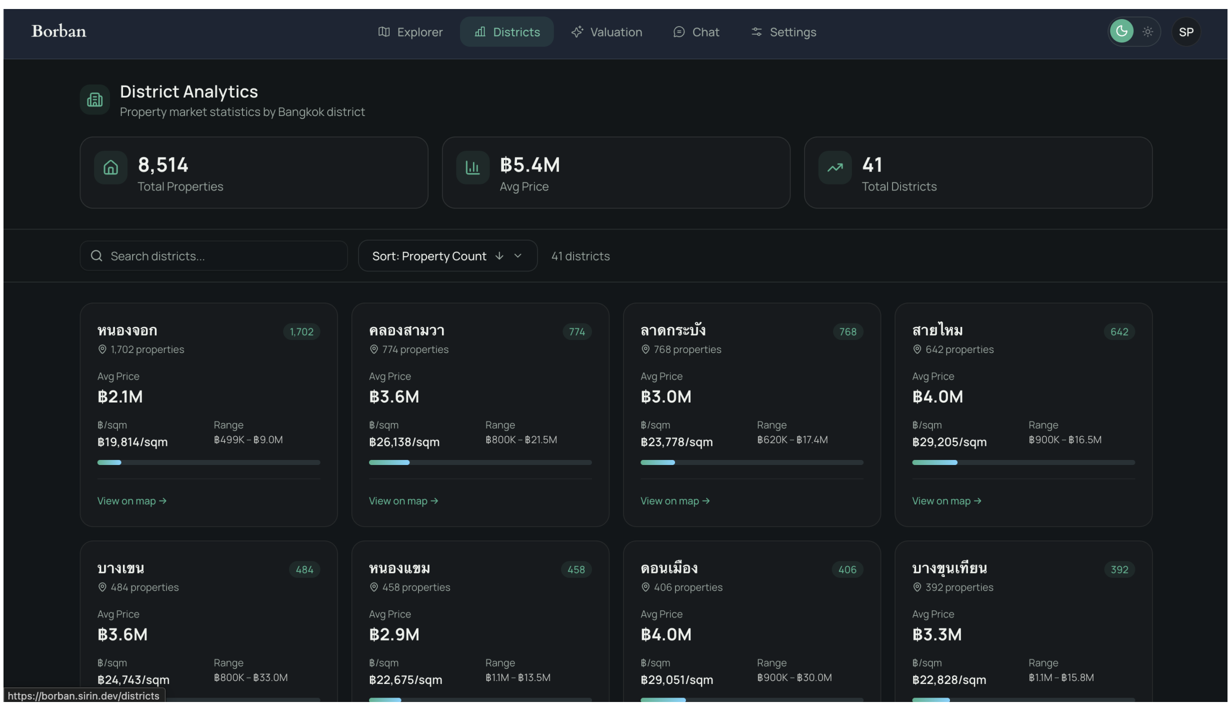
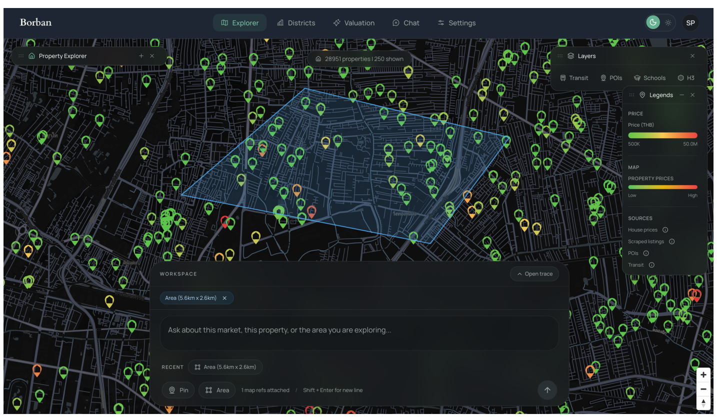
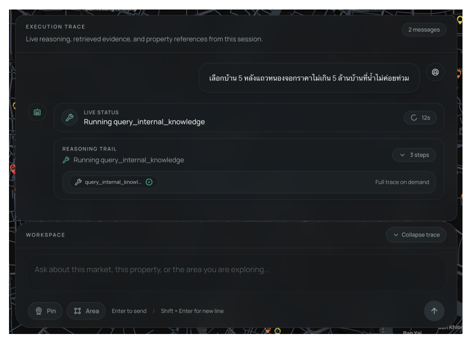


Figure 7.8: District Analytics Page



(a) Agent-assisted map workspace



(b) Chat-focused agent interface

Figure 7.9: AI Agent Interfaces

7.3 Technical and System Deliverables

In addition to the frontend pages, the project delivers the technical infrastructure required to make the platform operational. These outputs are important because BorBann depends on geospatial queries, AI inference, retrieval support, and operational controls rather than static page rendering alone.

Table 7.3 lists the main backend and operational deliverables. These items are the runtime foundation that allows the frontend workflows to access data, compute analytics, and serve AI-assisted responses.

Technical Deliverable	**Implementation Form**	**Delivered Function**
Geospatial backend API	FastAPI services	Exposes valuation, chat, site analysis, location intelligence, listing, and district analytics endpoints.
Spatial database platform	PostgreSQL + PostGIS	Stores and queries property, POI, transit, and district-level geographic data.
Retrieval layer	pgvector and internal knowledge services	Supports evidence-based chatbot responses and location-specific retrieval.
Valuation engine	Production LightGBM path	Returns predicted price, confidence, explanations, district comparison, and H3 context.
Explainability support	SHAP and feature-importance pipeline	Produces local and global factor explanation paths with measurable evaluation support.
Observability package	Metrics and refresh monitoring	Prometheus-compatible metrics, cache indicators, refresh counters, and health signals.
Deployment package	Docker Compose and runbook	Production stack definition, deployment scripts, environment guidance, and rollback support.
CI/CD package	GitHub Actions workflows	Automated test, lint, frontend build, and quality-gate execution.

Table 7.3: Technical Deliverables

7.4 Testing and Evaluation Report

The project deliverables include detailed testing and evaluation outputs. These are necessary because BorBann contains multiple layers of functionality: user interface workflows, backend APIs, geospatial analytics, machine learning prediction, and AI-agent behavior.

Testing Coverage Summary

Table 7.4 outlines the validation methodologies implemented in the system. It combines automated correctness checks, build validation, agent quality gates, and load-testing evidence.

Test Area	Representative Coverage	Purpose of Validation
Backend unit tests	13 unit test modules covering predictor types, feature contributions, explainability utilities, internal knowledge, and tool planning behavior.	Verify correctness of core service logic before integration.
Backend integration tests	2 integration modules covering chat status, provider validation, SSE event flow, session handling, attachments, and event ordering.	Confirm that API contracts and end-to-end chat flows work correctly.
Frontend validation	CI frontend job runs dependency install, lint, test, and production build.	Ensure the UI architecture remains robust and complies with project quality standards.
Quality gate checks	Dedicated CI quality-gate job with a 10-case agent benchmark policy.	Prevent unreviewed degradation in chatbot quality when a benchmark report exists.
Load validation	k6 Sprint A reports and optimization follow-up reports.	Measure latency, error rate, cache behavior, and operational bottlenecks under load.

Table 7.4: Testing Coverage Summary

The evaluation phase yielded empirical performance metrics rather than isolated test definitions. The next two tables separate automated validation evidence from performance evidence so that each result remains readable and fits the thesis layout cleanly.

Automated Validation Result

Table 7.5 lists the primary quality assurance checks integrated into the continuous integration pipeline.

Evaluation Item	Method	Observed Result	Interpretation
CI backend validation	GitHub Actions unit and integration jobs	Automated backend tests run with Python 3.13 and PostGIS service support.	The system delivers a repeatable backend validation flow for both isolated logic and API-level behavior.
CI frontend validation	GitHub Actions frontend job	Frontend pipeline runs dependency install, lint, test, and production build.	The frontend architecture operates as a continuously validated application to ensure robustness.
Chat API contract tests	Integration testing of chat endpoints	Verified SSE content type, session ID return, attachment handling, and event ordering such as thinking-first and done-last behavior.	The chat system includes explicit verification of stream behavior and session-based interaction.
Valuation service tests	Unit testing of prediction components	Baseline predictor types, distance calculations, dataclasses, and predictor registry behavior are tested.	Core valuation service behavior is guarded by repeatable tests before model output is used by the UI.
Golden benchmark quality gate	Ten-case benchmark with score and tool-first requirements	The benchmark requires 100% tool-first compliance for factual queries and a median judge score of at least 7.0 across 10 cases.	This gives the team a measurable way to identify chatbot weaknesses rather than relying on anecdotal review.

Table 7.5: Automated Validation Results

Performance and Load Result

Table 7.6 reports the measured performance figures captured during Sprint A load testing and the later tile optimization cycle.

The testing results show two important points. First, BorBann has delivered validation artifacts across functional correctness, interface behavior, and performance. Second, the reports do not hide remaining weaknesses. The chatbot benchmark still requires quality recovery, and some geospatial-heavy workflows remain above the desired latency target in local stress conditions.

7.5 Model Evaluation Report

The valuation component is one of the main deliverables of the project, so its report must include not only prediction output but also model-selection evidence. The project currently uses the LightGBM-based C5 path as the stable production baseline because it remains more reliable than later C6-family candidates under the accepted benchmark policy.

Grouped Benchmark Result Table

Table 7.7 reports the grouped benchmark results from the canonical benchmark summary. These values show the absolute error and fit metrics used to compare the stable C5 path with later C6-family candidates.

Table 7.8 focuses on the percentage regression against the C5 baseline. These relative values make it easier to explain why later models were not promoted, even when they added more complexity.

Table 7.9 adds the time-split check and the extra XGBoost comparison rows, which together complete the 12 canonical benchmark entries evaluated in the study.

Model Selection Interpretation

Table 7.10 converts the benchmark evidence into a deployment decision. It shows why C5 remains the production baseline even though more experimental variants were also implemented and benchmarked.

Explainability Evaluation Summary

The project also delivers model-explanation support beyond prediction accuracy. The explainability evaluation code reports seven quantitative signals: one SHAP rank-correlation value, two overlap measures (top-5 and top-10), three perturbation-based faithfulness measures, and one expected-feature coverage value. These outputs are important because they provide evidence that the explanation layer is tied to model behavior rather than only visual presentation.

Table 7.11 summarizes the explainability evaluation package and the measurable checks included in the implementation.

Evaluation Item	Method	Observed Result	Interpretation
Load test baseline	k6 Sprint A Run F	HTTP request failure rate 0.1590, average latency 15635.48 ms, p95 latency 37167.72 ms, check success 1.0, and 4321 requests over the run.	Stability improved after transit-query and transaction-recovery fixes, but the local profile still misses the intended latency target.
Baseline cache behavior	Post-run observability snapshot for Run F	Listings tile cache hit rate 0.0591, location-intelligence cache hit rate 0.0, and database query errors reduced from 1093 in Run E to 0 in Run F.	The run confirms cleaner backend stability, but cache effectiveness remains uneven across expensive workflows.
Load optimization result	Materialized tile source comparison	Representative mixed-workload performance improved from 1.81% failed requests and 12.6 s p95 latency to 0.00% failed requests and 6.63 s p95 latency.	Tile-serving optimization clearly improved reliability and tail latency, even though some non-tile bottlenecks remain.
Tile endpoint spot validation	Functional validation after optimization	Cold tile response was about 46 ms, warm tile response about 2 ms, and refresh duration about 2.7–3.1 s with 1 successful and 0 failed refresh cycles observed.	The optimized tile path is significantly cheaper after caching and materialized-source support.

Table 7.6: Performance and Load Results

Dataset	Stage	Overall MAE	Treasury MAE	Listing MAE	R^2
clean_grouped	C5	1,006,223	802,621	4,385,310	0.779
clean_grouped	C6	1,594,915	1,297,796	6,526,043	0.454
clean_grouped	C6I	1,575,586	1,280,060	6,480,273	0.469
clean_grouped	C6IR	1,575,491	1,280,060	6,478,605	0.469
all_grouped	C5	1,822,393	816,535	3,334,910	0.819
all_grouped	C6	2,811,255	1,304,607	5,076,815	0.564
all_grouped	C6I	2,835,192	1,268,005	5,191,786	0.571
all_grouped	C6IR	2,835,223	1,268,005	5,191,864	0.571

Table 7.7: Grouped Model Benchmark Results

Stage	Clean Listing Delta	Clean Treasury Delta	All Listing Delta	All Treasury Delta
C5	0.00%	0.00%	0.00%	0.00%
C6	+48.82%	+61.70%	+52.23%	+59.77%
C6I	+47.78%	+59.48%	+55.67%	+55.29%
C6IR	+47.74%	+59.48%	+55.68%	+55.29%

Table 7.8: Regression Against the C5 Baseline

Dataset	Model	Stage	Overall MAE	Listing MAE	R^2
clean_time	LightGBM	C5	998,152	6,862,957	0.784
clean_time	LightGBM	C6	1,318,455	10,019,947	0.617
clean_grouped	XGBoost	C6	1,560,390	6,462,368	0.475
all_grouped	XGBoost	C6	2,800,799	5,128,357	0.573

Table 7.9: Time-Split and Additional Model Rows

Evaluation Topic	Observed Evidence	Decision
Baseline stability	C5 shows the best balance between listing accuracy and treasury stability in both clean and all-grouped settings.	Keep C5 as the production-ready model path.
Candidate model regression	C6-family variants regress strongly on listing and treasury MAE compared with C5.	Do not promote C6-family models to production.
Time-split robustness	On the clean time split, C6 remains materially worse than C5 in overall, treasury, and listing MAE.	Avoid replacing the baseline with a weaker time-split performer.
Complexity versus usefulness	More complex variants, including image-augmented branches and HGT research direction, do not currently deliver stronger benchmark results.	Keep advanced models as research outputs, not default deployment models.
Governance and acceptance policy	The system implements explicit acceptance gates based on listing improvement and treasury regression caps.	Model promotion is controlled by benchmark evidence rather than assumption.

Table 7.10: Model Selection Interpretation

The explainability evaluation was conducted using automated testing procedures to validate the model's interpretation metrics. Table 7.12 summarizes the performance metrics obtained during this evaluation phase.

Overall, the model evaluation deliverable is not limited to one best score. It includes benchmark tables, model-selection rules, and explanation-evaluation support so that the delivered valuation feature remains both useful and accountable.

Explainability Item	**Delivered Mechanism**	**Purpose**
Local explanation	Per-request local SHAP contribution output converted to approximate THB effect.	Show the main request-specific factors affecting the estimate.
Global reference explanation	Gain-based feature-importance summary from the model.	Provide stable model-level signal reference.
Rank agreement check	Spearman rank correlation plus top-5 and top-10 overlap between SHAP and gain rankings.	Verify whether different explanation views remain aligned on important features.
Faithfulness check	Perturbation delta for top-ranked features, random-reference delta, and faithfulness lift.	Test whether important features cause meaningful prediction change.
Expected-feature coverage	Coverage check for expected domain factors such as area and major distance features within the inspected top-k set.	Confirm that important domain features appear in explanation results.

Table 7.11: Explainability Evaluation Deliverables

Metric	**Current Result**	**Interpretation**
SHAP rank correlation	0.90	SHAP ranking is strongly aligned with the gain-based global ranking.
Top-5 overlap	0.80	Four of the top five important features are shared between the two ranking views.
Faithfulness lift	1.70×	Top SHAP-ranked features produce about 1.70 times more prediction change than random features.
Expected-feature coverage	1.00	All expected domain features in the tested report are present in the inspected top-ranked set.

Table 7.12: Explainability Evaluation Results

Chapter 8

Conclusion and Discussion

8.1 Conclusion

BorBann delivers a unified real estate information platform that combines data integration, geospatial analysis, explainable price prediction, and AI-assisted interaction in one system. The project addresses an important problem in the Thai real estate context, where useful information is often fragmented across separate platforms and difficult to compare. By bringing together listings, public records, location intelligence, and model-based valuation, the platform gives users a clearer basis for decision making.

From the implementation and evaluation results, the project demonstrates several practical outcomes. First, the system can support map-based property exploration together with localized indicators such as transit access, walkability, school access, flood risk, and noise exposure. Second, the valuation module provides not only an estimated price but also supporting signals that help users understand why the result is produced. Third, the AI chat workflow improves accessibility for non-technical users because they can ask questions in natural language instead of relying only on manual filtering.

The project also shows the importance of disciplined validation. Benchmark results indicate that the stable LightGBM baseline remains more suitable for deployment than more complex experimental candidates under the current data conditions. In the same way, chatbot quality and performance results demonstrate that AI features must be monitored with explicit tests and quality gates rather than assumed to be correct.

User Feedback

To collect early feedback, the team distributed a Google Form after demonstrating the system to three users. The goal of this activity was not to claim a large usability study, but to gather initial reactions from likely target users and identify practical improvement points.

Overall, the feedback was positive toward the system concept and the integration of multiple forms of analysis in one interface. The responses also confirmed that trust, explanation clarity, and chatbot accuracy should remain high-priority improvement areas.

User	Profile	Feedback Summary
User 1	First-time homebuyer	Found the map and valuation summary easy to understand. Requested clearer labels for some indicators before making a final decision.
User 2	First-time homebuyer	Considered the location intelligence panel helpful when comparing neighborhoods. Wanted a clearer explanation of confidence and risk colors.
User 3	Real estate investor	Valued the district comparison and localized risk information. Wanted stronger confidence guidance when comparing several properties.

Table 8.1: Preliminary Google Form Feedback from Three Users

8.2 Discussion

Key Areas for Future Focus

The next phase of the project should focus on a few areas that have the highest impact on user trust and system usefulness.

- **Source completeness and traceability:** Every indicator should have a complete source reference so that users can verify where the information comes from.
- **Chatbot retrieval quality:** Location-specific questions should continue to be improved through stronger retrieval, better benchmark coverage, and tighter grounding rules.
- **Performance for heavy geospatial workflows:** The platform should continue reducing latency for location intelligence and dense map interactions.
- **Model and data refresh cycle:** New transaction data, listing updates, and periodic retraining are needed to keep the valuation result current.
- **Broader user evaluation:** A larger user study would provide stronger evidence on usability, trust, and feature usefulness across different user groups.

Challenges Encountered

During development, the team faced several challenges that influenced both implementation effort and system quality.

- **Data fragmentation:** Real estate, transit, POI, and risk data come from different sources with different formats and update cycles.

- **Model selection trade-offs:** More advanced model variants did not automatically provide better real-world performance, so benchmark discipline was necessary.
- **Explainability requirements:** It was not enough to show factors visually; the project also needed methods to verify that those factors truly affected the prediction.
- **Grounded AI responses:** The chatbot needed stronger retrieval and tool-use controls to avoid vague or mismatched answers for specific neighborhoods.
- **System performance:** Combining geospatial analytics, AI inference, and interactive visualization created non-trivial latency and operational complexity.

In summary, the project achieved its main objective of building a practical and explainable real estate information platform for Bangkok. At the same time, the discussion points show that the most important future work is not only adding new features, but also strengthening trust, precision, and operational readiness.

Appendix A

Calculation Formula Reference

This appendix documents the main calculation logic used by the delivered BorBann analytics components. It starts with the location intelligence formulas because these scores appear throughout the property detail, valuation, and agent-assisted workflows.

A.1 Location Intelligence Composite Score

The location intelligence service combines five component scores into one composite score on a 0–100 scale. According to the implementation in the backend location intelligence service, the composite score is computed as a weighted sum:

$$\text{Composite Score} = 0.25T + 0.25W + 0.20S + 0.15F + 0.15N \tag{A.1}$$

where:

- T is the transit score,
- W is the walkability score,
- S is the schools score,
- F is the normalized flood score,
- N is the normalized noise score.

The implementation converts the final result to an integer before returning it in the API response.

Table A.1 summarizes the exact factor weights used in the composite calculation.

Component	Weight
Transit score	0.25
Walkability score	0.25
Schools score	0.20
Flood score	0.15
Noise score	0.15

Table A.1: Location Intelligence Composite Weights

A.2 Transit Score Formula

The transit score is an additive score capped at 100 points. It is based on three sources of transport accessibility: nearest rail access, number of bus stops within 500 meters, and optional ferry access.

$$T = \min(100, T_{rail} + T_{bus} + T_{ferry}) \tag{A.2}$$

with component rules defined as follows:

$$T_{rail} = \begin{cases} 40, & d_{rail} \leq 400 \\ 30, & 400 < d_{rail} \leq 800 \\ 20, & 800 < d_{rail} \leq 1000 \\ 0, & \text{otherwise} \end{cases} \tag{A.3}$$

$$T_{bus} = \begin{cases} 30, & b \geq 5 \\ 20, & 3 \leq b < 5 \\ 10, & 1 \leq b < 3 \\ 0, & b = 0 \end{cases} \tag{A.4}$$

$$T_{ferry} = \begin{cases} 20, & d_{ferry} \leq 500 \\ 10, & 500 < d_{ferry} \leq 1000 \\ 0, & \text{otherwise} \end{cases} \tag{A.5}$$

where d_{rail} is the distance in meters to the nearest rail station, b is the number of bus stops within 500 meters, and d_{ferry} is the distance in meters to the nearest ferry pier.

A.3 Schools Score Formula

The schools score is based on the number of schools found within 2 kilometers, with an additional bonus for educational variety.

$$S = \min(100, S_{count} + S_{variety}) \tag{A.6}$$

The count-based component is:

$$S_{count} = \begin{cases} 80, & n \geq 10 \\ 60, & 5 \leq n < 10 \\ 40, & 2 \leq n < 5 \\ 20, & n = 1 \\ 0, & n = 0 \end{cases} \tag{A.7}$$

The variety bonus is:

$$S_{variety} = \begin{cases} 20, & v \geq 3 \\ 10, & v = 2 \\ 0, & v \leq 1 \end{cases} \tag{A.8}$$

where n is the total number of schools within 2 kilometers and v is the number of detected school-level categories.

A.4 Walkability Score Formula

The walkability score evaluates eight amenity categories within 800 meters: restaurant, cafe, grocery, pharmacy, bank, park, gym, and retail. Each category contributes points based on whether at least one or at least three amenities are present.

$$W = \min\left(100, \sum_{i=1}^{8} w_i\right) \tag{A.9}$$

For each category i:

$$w_i = \begin{cases} 12.5, & c_i \geq 3 \\ 8, & 1 \leq c_i < 3 \\ 0, & c_i = 0 \end{cases} \tag{A.10}$$

where c_i is the number of amenities found in category i.

A.5 Flood and Noise Normalization

Flood and noise are first classified qualitatively and then converted into normalized numerical values for the composite score.

The flood normalization used in the implementation is:

$$F = \begin{cases} 100, & \text{low flood risk} \\ 50, & \text{medium flood risk} \\ 20, & \text{high flood risk} \\ 50, & \text{unknown} \end{cases} \tag{A.11}$$

The noise normalization used in the implementation is:

$$N = \begin{cases} 100, & \text{quiet} \\ 70, & \text{moderate} \\ 40, & \text{busy} \\ 50, & \text{unknown} \end{cases} \tag{A.12}$$

The noise class itself is determined from the minimum detected distance to the major-road proxy used by the service.

$$\text{Noise Level} = \begin{cases} \text{busy}, & d < 100 \\ \text{moderate}, & 100 \leq d < 300 \\ \text{quiet}, & d \geq 300 \end{cases} \tag{A.13}$$

where d is the minimum available distance in meters to the road-noise proxy source.

These formulas explain how BorBann transforms raw geospatial evidence into a single interpretable location score that can be shown in the UI and reused by other workflows such as valuation and agent-assisted property analysis.

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